

Hellenic Complex Systems Laboratory

Uncertainty Estimation of Diagnostic Accuracy Measures under Parametric Distributions

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Uncertainty Estimation of Diagnostic Accuracy Measures under Parametric Distributions

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Abstract

Background: Diagnostic tests are evaluated by their ability to discriminate between diseased and nondiseased individuals; however, estimates of diagnostic accuracy alone do not quantify their associated uncertainty. Uncertainty assessment is therefore essential for interpreting accuracy estimates and comparing diagnostic thresholds.

Methods: We develop a parametric framework for propagating uncertainty to diagnostic accuracy measures formulated as functions of sensitivity, specificity, and prevalence. Normal, lognormal, or gamma models are specified separately for the diseased and nondiseased populations. At a fixed threshold, first-order uncertainty propagation retains covariance between estimated population means and standard deviations, incorporates prevalence sampling uncertainty when applicable, and represents calibration uncertainty through a shared diagnostic threshold. The Wolfram Language implementation provides uncertainty components by source, joint output covariance, and approximate expanded-uncertainty intervals calculated using effective degrees of freedom derived from the source-specific variance contributions. At optimized thresholds, shared-threshold uncertainty is propagated through the nonlinear measurement model by Gauss–Hermite quadrature, conditional on the estimated optimal threshold.

Results: Illustrative analyses indicate that the contribution of each uncertainty source varies with the diagnostic accuracy measure, distributional assumptions, and threshold. Dependence induced by the shared threshold can partially cancel its contribution to the uncertainty of a derived measure. In the cases examined, joint propagation reproduces the corresponding marginal variances and enables evaluation of uncertainty across diagnostic thresholds.

Conclusions: The framework provides a reproducible basis for assessing diagnostic accuracy and uncertainty while preserving relevant input and output dependencies. Decomposition by source helps identify the assumptions and uncertainty components underlying each estimate. The intervals remain local approximations; nominal coverage and additional uncertainty from threshold selection require separate assessment.

Keywords:

diagnostic accuracy; covariance propagation; measurement uncertainty; sampling uncertainty; diagnostic threshold; delta method; lognormal distribution; effective degrees of freedom; Wolfram Language

1. Introduction

Medical diagnosis integrates clinical evidence with a model of disease status [1]. Quantitative tests classify measurements relative to a diagnostic threshold. Their performance depends on the measurement distributions in diseased and nondiseased populations and on the population in which the test is applied [2–4].

Diseased and nondiseased populations are denoted by D and H. Throughout, results are positive when $X \geq t$ and negative when $X < t$. Figure 1 illustrates overlapping population distributions; Table 1 defines the corresponding contingency table.

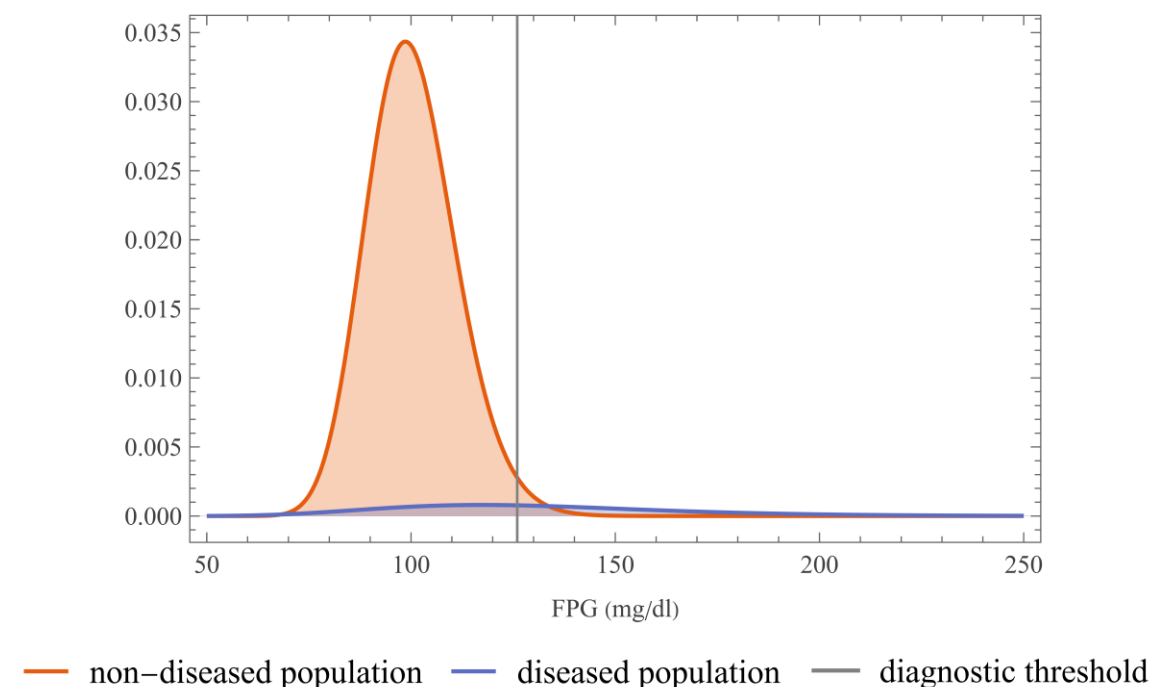


Figure 1. Lognormal probability densities for the supplied FPG inputs at an illustrative threshold of 126 mg/dL.

Table 1. Contingency table for a positive-above-threshold test.

Test result	Diseased (D)	Nondiseased (H)
Positive ($X \geq t$)	TP	FP
Negative ($X < t$)	FN	TN

1.1.Diagnostic accuracy measures

The sixteen measures are sensitivity (Se), specificity (Sp), overall diagnostic accuracy (ACC), positive and negative predictive values (PPV, NPV), diagnostic odds ratio (DOR), positive and negative likelihood ratios (LR^+ , LR^-), Youden’s index (J), balanced accuracy (BA), concordance probability (CZ), Fowlkes–Mallows index (FMI), Cohen’s kappa (κ), prevalence- and bias-adjusted kappa (PABAK), F_1 score (F_1), and Matthews correlation coefficient (MCC) [4, 5].

Table 2. Interpretation and admissible range of the sixteen diagnostic accuracy measures.

Measure	Interpretation	Range
Se	Proportion positive among diseased individuals.	$[0, 1]$
Sp	Proportion negative among nondiseased individuals.	$[0, 1]$
ACC	Prevalence-weighted probability of correct classification.	$[0, 1]$
PPV	Probability of disease conditional on a positive result.	$[0, 1]$
NPV	Probability of no disease conditional on a negative result.	$[0, 1]$
DOR	Ratio of the odds of a positive result in the two populations.	$[0, \infty)$
LR ⁺	Se divided by the false-positive probability.	$[0, \infty)$
LR ⁻	False-negative probability divided by Sp.	$[0, \infty)$
J	Se plus Sp minus one.	$[-1, 1]$
BA	Arithmetic mean of Se and Sp.	$[0, 1]$
CZ	Product of Se and Sp.	$[0, 1]$
FMI	Geometric mean of Se and PPV.	$[0, 1]$
κ	Agreement corrected for the chance agreement from both marginals.	$[-1, 1]$
PABAK	$2 \text{ ACC} - 1$; generally depends on prevalence.	$[-1, 1]$
F_1	Harmonic mean of Se and PPV.	$[0, 1]$
MCC	Correlation between binary test classification and disease status.	$[-1, 1]$

Increasing the threshold decreases Se and increases Sp under this decision rule. This trade-off between detection and false positives does not alone determine a clinically optimal threshold. PPV and NPV also depend on prevalence, and operational decisions may depend on consequences and costs [3].

1.2. Uncertainty

Standard uncertainty expresses uncertainty on a standard-deviation scale. For a derived measure, specified input uncertainties and covariances are propagated through the measurement model [6, 7]. DAM uncertainty is therefore conditional on the model, sampling design, and represented uncertainty sources.

Parameter and prevalence sampling uncertainties constitute total sampling uncertainty. Combined standard uncertainty additionally includes shared-threshold measurement uncertainty.

The calibration model represents analytical uncertainty through one effective decision threshold shared by both populations. Its contribution is evaluated by applying the same threshold shift to both distribution tails while holding the fitted distributions fixed; parameter and prevalence sampling uncertainties are evaluated separately. An upward shift decreases Se and increases Sp, with the reverse occurring for a downward shift. This common source induces negative covariance between the two rates, which must be retained when propagating uncertainty to measures that depend on both [6].

Independent perturbations of Se and Sp assign separate uncertainty sources and omit their shared-threshold dependence. Such a model requires separate justification, uncertainty magnitudes, and

dependence assumptions. Even with identical marginal uncertainties, it generally yields different uncertainties for measures combining the two rates. The present model instead propagates their joint response to a shared threshold, conditional on the specified population distributions.

Random analytical variation in individual measurements can alter the observed population distributions and inferred diagnostic performance. Under an independent additive model, each observed distribution is the convolution of the underlying population distribution with the distribution of analytical variation. This may affect distribution tails, the ROC curve, and the estimated optimal threshold. Analysis requires an explicit model specifying the mechanism, magnitude, and dependence structure of analytical variation. It must also distinguish underlying distributions that exclude analytical variation from observed distributions that include it, thereby preventing double counting of variation already represented in the observed dispersion [8–10].

1.3.Measurement distributions

Each population is modeled independently as normal, lognormal, or gamma using arithmetic-scale means and standard deviations. The selected family determines the distribution tails and the third and fourth central moments used in sampling covariance, affecting both point estimates and uncertainty [11–13].

1.4.Previous research

Parametric ROC analysis and analytical measurement uncertainty offer complementary perspectives on diagnostic performance [8–10, 13–16]. Threshold-specific uncertainty analysis connects these perspectives through explicitly defined primitive inputs and dependence assumptions.

Previous tools examined relationships between diagnostic accuracy and measurement uncertainty, diagnostic-measure uncertainty, and Bayesian diagnostic probabilities [17–19]. The present framework uses an implementation that explicitly accounts for covariance among the input estimates. The framework retains within-population mean–standard deviation covariance and dependence induced by a shared threshold. Its source decomposition follows the covariance form of the law of propagation of uncertainty [17–19].

2. Methods

2.1.Study inputs and diagnostic performance

We modeled test measurements separately in diseased and nondiseased groups, using a normal, lognormal, or gamma distribution for each group. The inputs were each group’s sample size, arithmetic mean, and standard deviation, together with disease prevalence and the uncertainty in the diagnostic threshold. A result at or above the threshold was classified as positive. The fitted distributions gave Se and Sp; these values and prevalence were then used to calculate the sixteen measures listed in Table 2.

2.2.Sources of uncertainty

We included uncertainty arising from the finite samples, from prevalence when applicable, and from a common shift in the diagnostic threshold [6]. The calculation retained the relationship between the estimated mean and standard deviation within each group. A threshold shift was applied to both groups together: raising the threshold lowers Se and raises Sp. Accounting for this shared response is necessary when calculating uncertainty in measures that combine both rates.

Threshold uncertainty was treated as a shared calibration effect and evaluated separately from random analytical variation in individual test measurements. In the illustrative analysis, the approximate population distributions were adjusted to remove the mean contribution of

measurement variance before threshold uncertainty was incorporated, thereby avoiding double counting of analytical variation.

2.3. Prevalence

For the illustrative cohort calculation, prevalence was the proportion classified as diseased, with binomial sampling uncertainty. In a case-control study with deliberately fixed group sizes, that proportion need not represent clinical prevalence.

2.4. Uncertainty estimates and intervals

At each specified threshold, uncertainty was propagated using a first-order approximation, yielding source-specific contributions, the combined standard uncertainty, and an approximate pointwise expanded-uncertainty interval at a specified nominal coverage level [6, 20, 21]. Standard uncertainty is expressed on a standard-deviation scale; the interval gives a wider range around the estimate. Intervals depend on the fitted model and may be less reliable near the limits of a measure or when a ratio has a very small denominator. Restricting an interval to its admissible range does not establish its coverage.

2.5. Threshold comparison and implementation

We evaluated performance and uncertainty across a threshold range. For the optimization analysis, the software selected the threshold that maximized each chosen measure above the nondiseased mean. At that selected threshold, it used nonlinear propagation for the shared-threshold measurement uncertainty. These intervals quantify uncertainty in the diagnostic accuracy measure conditional on the estimated optimal threshold; they exclude uncertainty associated with threshold estimation and selection. A clinically preferred threshold also depends on the consequences of false-positive and false-negative results.

Calculations were performed in the *DiagAccU*, Version 2.0.0, Wolfram Language notebook. The case-study inputs and settings are reported in Tables 3 and 4. *Appendix A* gives the formulas, assumptions, derivatives, uncertainty calculations, and general calculation procedures. *Appendix B* lists notation and software information.

3. Results

3.1. Illustrative case study

The illustrative analysis uses fasting plasma glucose (FPG) data from 414 participants aged 65–67 years in NHANES 2005–2016 [27]. Reference classification by 2-h oral glucose tolerance test (OGTT) identified 52 participants with values of at least 200 mg/dL (diseased) and 362 with values below 200 mg/dL (nondiseased). The analysis is unweighted and does not provide a population estimate.

The reference-positive fraction and its binomial standard uncertainty are:

$$v = \frac{52}{414} = 0.125603865, \quad u_v = 0.016287538$$

Approximate lognormal distributions, with measurement uncertainty treated as negligible, were obtained by subtracting the mean measurement variance from each maximum-likelihood fitted variance. Table 3 gives the rounded, approximate parameters used in the uncertainty calculations.

Table 3. Rounded population inputs and lognormal summaries for the illustrative example. Concentrations are in mg/dL.

Analysis input or model quantity	Diseased	Nondiseased
Sample size	52	362
Analysis mean (mg/dL)	136.0	101.0
Analysis SD (mg/dL)	36.6	11.0
Model median (mg/dL)	131.327471809	100.406266869
Model skewness	0.826843601	0.328024529
Model kurtosis (not excess)	4.239774961	3.191905805
Log mean	4.877693989	4.609224624
Log SD	0.264427294	0.108589986

The lognormal distributions specified by the rounded arithmetic means and standard deviations are:

$$X_D \sim \text{Lognormal}(4.877693989, 0.264427294)$$

$$X_H \sim \text{Lognormal}(4.609224624, 0.108589986)$$

Uncertainty in the shared diagnostic threshold is propagated separately using the calibration model:

$$u_t = \sqrt{0.812^2 + (0.0119t)^2}$$

This uncertainty represents a shared shift of the decision boundary, distinct from the measurement variation removed from the approximate population distributions.

At the clinical threshold of 126 mg/dL, the standard uncertainty is:

$$u_t(126) = 1.705152298 \text{ mg/dL}$$

Table 4. Calculation settings for the illustrative example.

Setting	Value and scope
Population families	Lognormal-lognormal; alternative distribution families are compared in Figure 9.
Sample sizes	$n_D = 52$; $n_H = 362$
Arithmetic means	$m_D = 136.0$; $m_H = 101.0$ mg/dL
Arithmetic SDs	$s_D = 36.6$; $s_H = 11.0$ mg/dL
Threshold	$t = 126$ mg/dL for Figures 1, 4, 9, 10; 91–173 mg/dL for Figures 3, 5–8
Prevalence	$v = 52/414$; varied from 0.001 to 0.999 only in Figure 4
Threshold coefficients	$b_0 = 0.812$ mg/dL; $b_1 = 0.0119$
Prevalence sampling	Enabled.
Sampling degrees of freedom	$\nu_D = 51$; $\nu_H = 361$; $\nu_v = 413$
Threshold degrees of freedom	$\nu_t = \infty$
Intervals	Nominal two-sided 95%; Student's t coverage factor with source-block degrees of freedom; admissible-range restriction

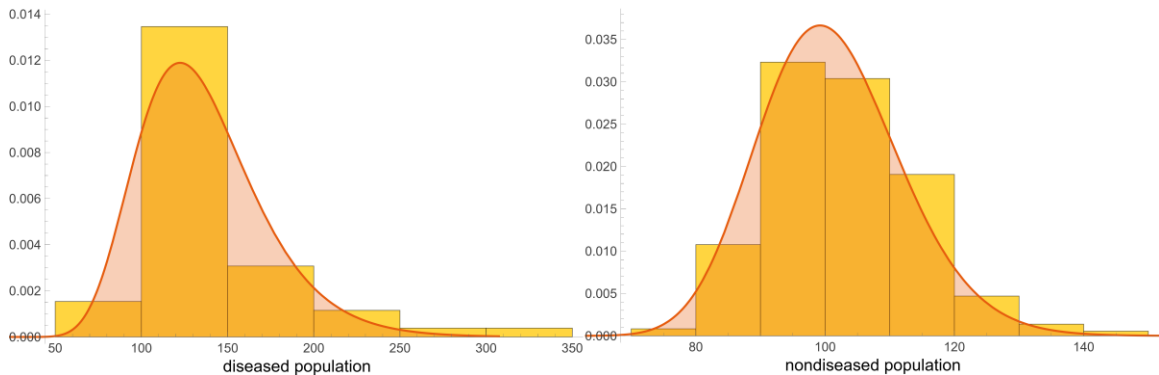


Figure 2. FPG histograms (yellow) and approximate lognormal distributions (orange) for 414 NHANES participants aged 65–67 years: 52 with 2-h OGTT values of at least 200 mg/dL (diseased) and 362 with values below 200 mg/dL (nondiseased). FPG is in mg/dL.

Figure 1 illustrates classification at 126 mg/dL, with FPG values at or above this threshold classified as positive.

3.2. Program application

The measures and their uncertainties are evaluated over 91–173 mg/dL using the settings in Table 4. Figure 4 examines prevalence effects at a fixed threshold.

3.3. Threshold-dependent profiles

Increasing the threshold reduces Se and increases Sp and, generally, PPV. NPV exhibits a shallow maximum before declining, whereas ACC peaks near 130 mg/dL. The measures therefore favor different thresholds (Figure 3).

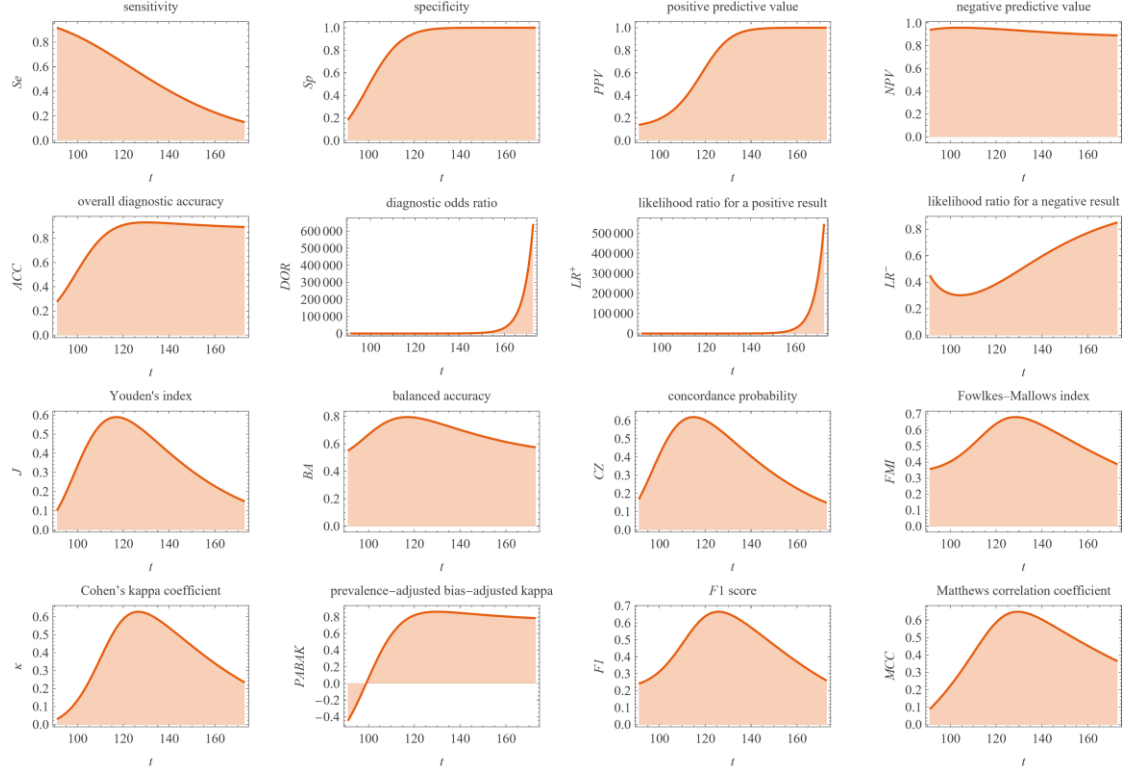


Figure 3. Profiles of the 16 diagnostic accuracy measures over thresholds of 91–173 mg/dL, using the population inputs and calibration settings in Tables 3 and 4.

3.4. Prevalence-dependent profiles

At 126 mg/dL, increasing prevalence raises PPV and lowers NPV. ACC and PABAK decrease, whereas FMI and F_1 increase. κ and MCC attain interior maxima (Figure 4).

The profiles show point estimates with fixed population distributions; prevalence-independent measures are omitted.

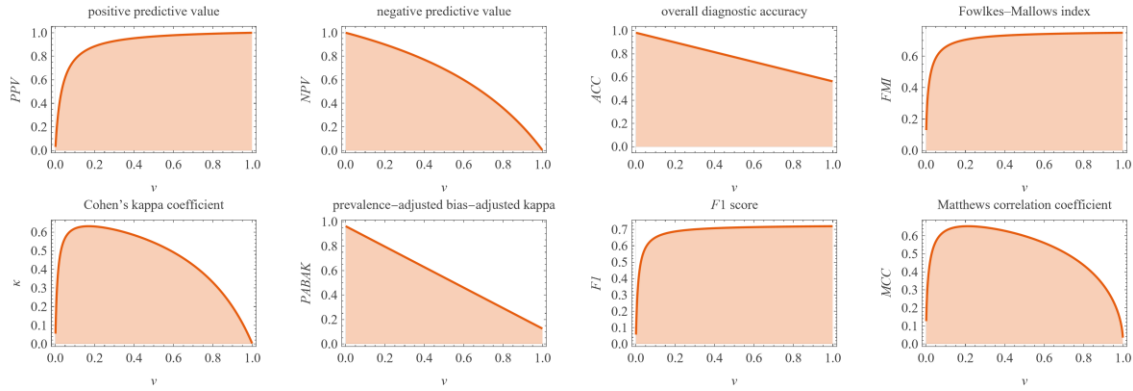


Figure 4. Profiles of eight diagnostic accuracy measures over prevalence values of 0.001–0.999, with the FPG threshold fixed at 126 mg/dL and the population distributions held constant.

3.5. Relations among measures

Figure 5 relates selected pairs of measures evaluated at a common threshold. Direction reversals indicate that one value of a measure may correspond to multiple values of another.

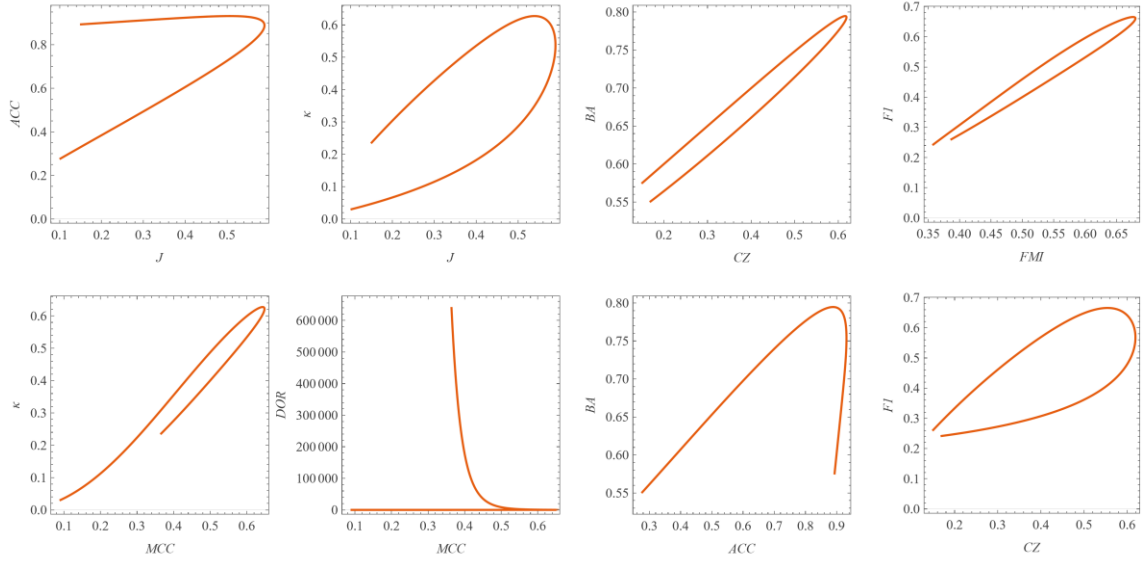


Figure 5. Relations among diagnostic accuracy measures over thresholds of 91–173 mg/dL. Each curve pairs measures evaluated at a common threshold. CZ denotes the Se–Sp product.

3.6. Uncertainty

Figure 6 presents total sampling, shared-threshold measurement, and combined standard uncertainties, with covariance retained in the propagation.

At 126 mg/dL, PPV has total sampling and shared-threshold measurement standard uncertainties of 0.0441 and 0.0406, respectively, and a combined standard uncertainty of 0.0600.

Sampling uncertainty dominates many measures near 126 mg/dL, whereas shared-threshold measurement uncertainty predominates for Sp and LR⁺. Cancellation between Se and Sp responses can reduce the first-order threshold contribution.

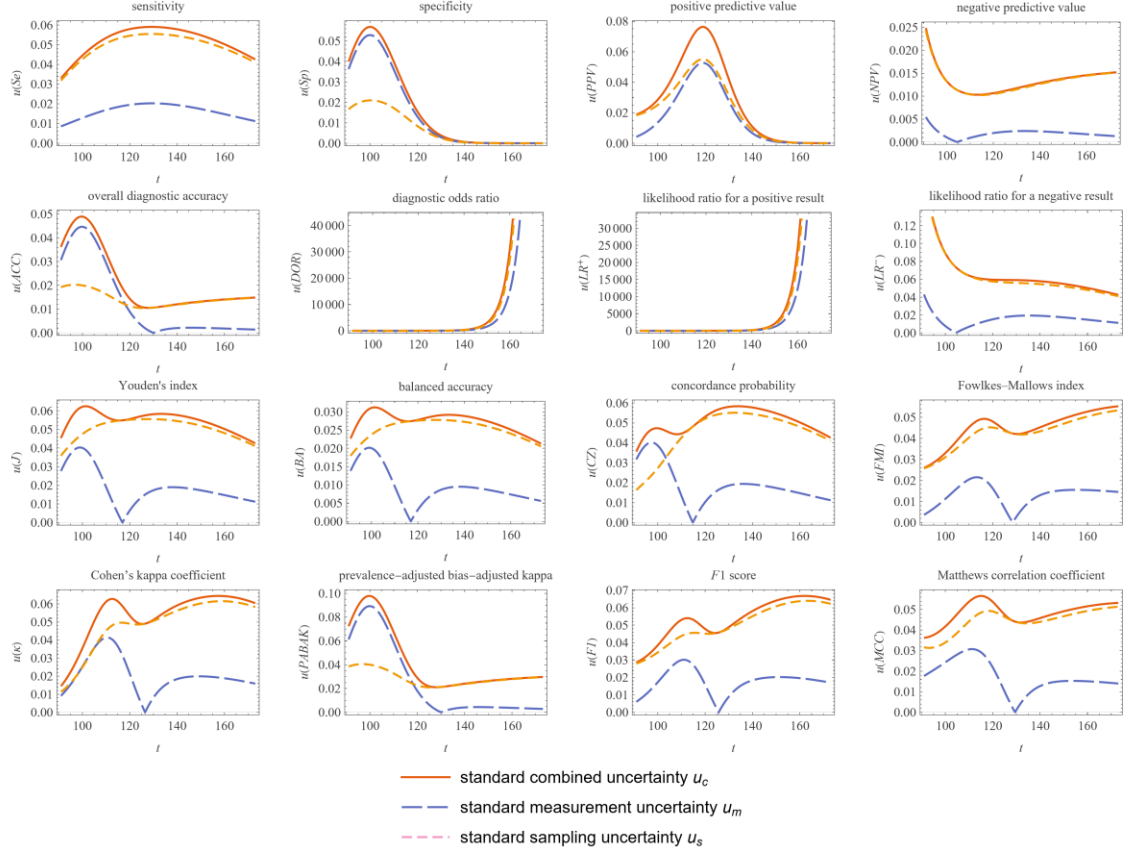


Figure 6. Combined (solid orange), shared-threshold measurement (long-dashed blue), and total sampling (short-dashed yellow-orange) standard uncertainties across thresholds. Sampling includes both population blocks and the enabled prevalence component; covariance is retained within each source.

3.7. Relative uncertainty

Relative standard uncertainty is the standard uncertainty divided by the absolute measure value. It can become very large near zero and is undefined at zero; neither feature independently indicates a change in diagnostic performance.

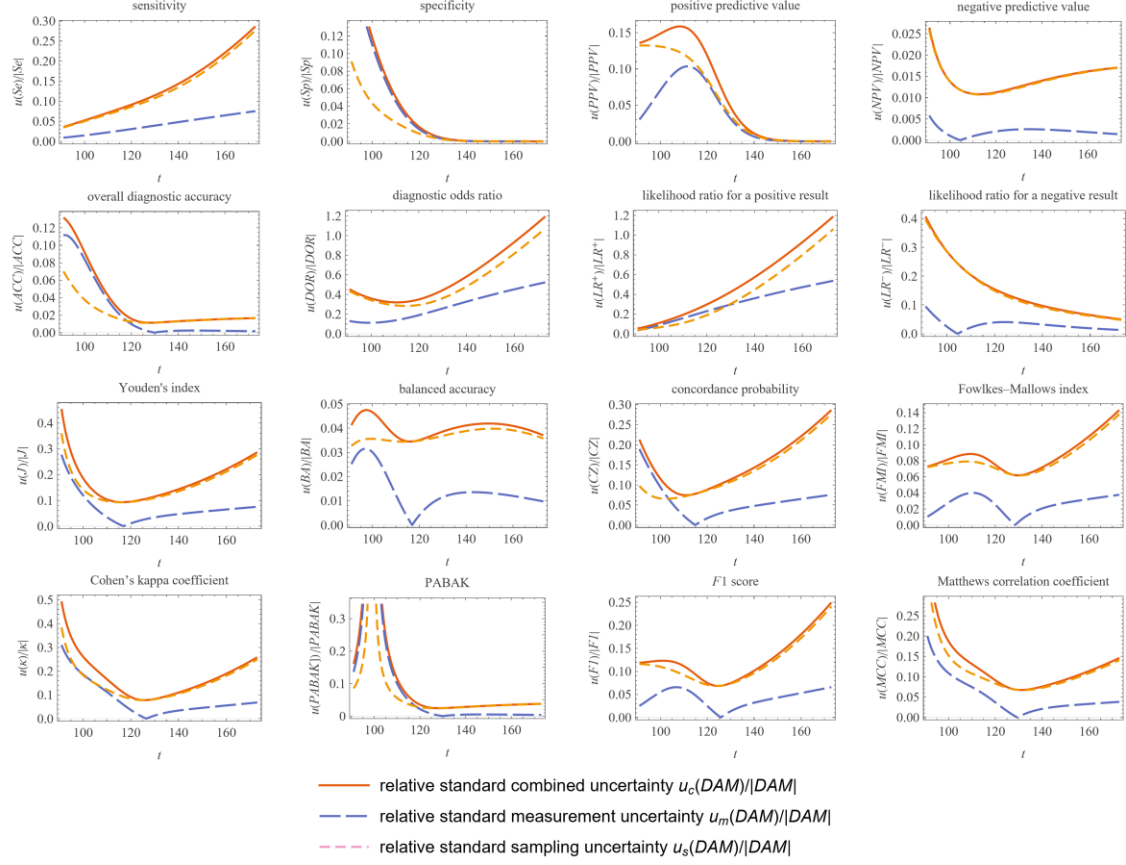


Figure 7. Relative standard uncertainties of the diagnostic accuracy measures, using the component styles in Figure 6. Relative uncertainty is undefined at zero measure values.

3.8. Pointwise uncertainty intervals

Figure 8 shows approximate pointwise 95% expanded-uncertainty intervals calculated using effective degrees of freedom derived from the source-specific variance contributions.. Interval limits are restricted to each measure's admissible range. The nominal 95% level refers to each fixed

threshold individually; these intervals do not constitute simultaneous confidence bands across thresholds.

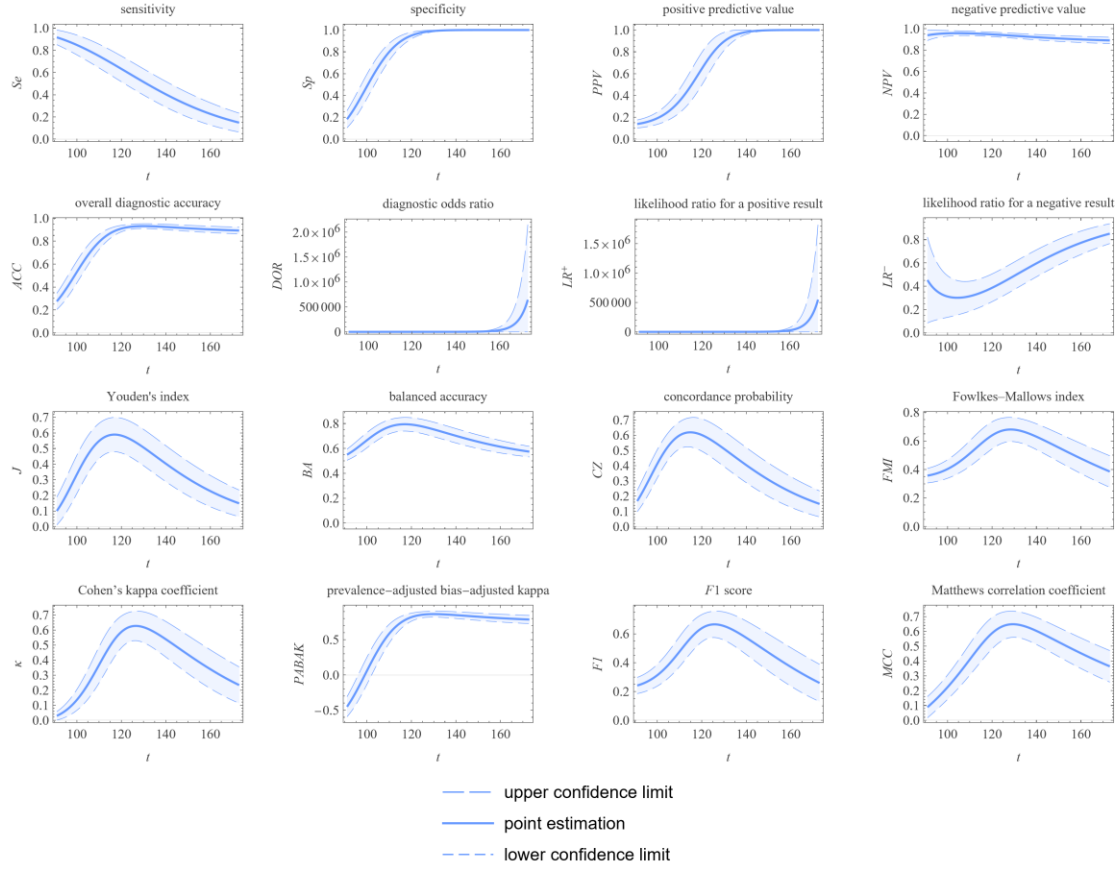


Figure 8. Point estimates (curves) and approximate pointwise 95% expanded-uncertainty intervals (shading and dashed boundaries) across thresholds.

3.9. Estimates and intervals at the clinical threshold

At 126 mg/dL, PPV is 0.8155, with standard uncertainty 0.0600 and an approximate 95% expanded-uncertainty interval of 0.6980–0.9331. NPV is 0.9398, with standard uncertainty 0.0113 and an approximate 95% expanded-uncertainty interval of 0.9176–0.9620. Figure 9 compares distribution families; Figure 10 presents estimates and intervals for all 16 measures under the lognormal model.

Measure	N / N	N / LN	N / G	LN / N	LN / LN	LN / G	G / N	G / LN	G / G
Se	0.6077	0.6077	0.6077	0.5622	0.5622	0.5622	0.5753	0.5753	0.5753
Sp	0.9885	0.9817	0.9840	0.9885	0.9817	0.9840	0.9885	0.9817	0.9840
ACC	0.9406	0.9347	0.9367	0.9349	0.9290	0.9310	0.9366	0.9307	0.9326
PPV	0.8834	0.8270	0.8450	0.8752	0.8155	0.8345	0.8776	0.8190	0.8376
NPV	0.9461	0.9457	0.9458	0.9402	0.9398	0.9399	0.9419	0.9415	0.9416
DOR	132.8809	83.2437	95.1461	110.1851	69.0259	78.8953	116.2051	72.7971	83.2058
LR ⁺	52.7422	33.2675	37.9373	48.7986	30.7801	35.1007	49.9310	31.4943	35.9152
LR ⁻	0.3969	0.3996	0.3987	0.4429	0.4459	0.4449	0.4297	0.4326	0.4316
J	0.5961	0.5894	0.5916	0.5507	0.5440	0.5462	0.5637	0.5570	0.5593
BA	0.7981	0.7947	0.7958	0.7754	0.7720	0.7731	0.7819	0.7785	0.7796
CZ	0.6007	0.5966	0.5979	0.5557	0.5520	0.5532	0.5686	0.5648	0.5661
FMI	0.7327	0.7089	0.7165	0.7014	0.6771	0.6850	0.7105	0.6864	0.6942
κ	0.6881	0.6649	0.6725	0.6503	0.6274	0.6349	0.6613	0.6383	0.6459
PABAK	0.8813	0.8695	0.8734	0.8699	0.8581	0.8620	0.8732	0.8614	0.8653
F ₁	0.7200	0.7005	0.7069	0.6846	0.6656	0.6718	0.6950	0.6758	0.6821
MCC	0.7032	0.6748	0.6840	0.6701	0.6410	0.6504	0.6797	0.6508	0.6602

Figure 9. Diagnostic accuracy estimates at 126 mg/dL for nine combinations of normal (N), lognormal (LN), and gamma (G) population distributions, with arithmetic means, standard deviations, sample sizes, and prevalence held fixed.

Measure	Estimate	u_c	V_{eff}	k	Lower	Upper
Se	0.562224	0.058942	65.423	1.996893	0.444523	0.679925
Sp	0.981734	0.007023	2675.097	1.960851	0.967963	0.995506
ACC	0.929042	0.010699	253.283	1.969374	0.907972	0.950112
PPV	0.815547	0.059962	2256.318	1.961016	0.697961	0.933134
NPV	0.939801	0.011265	253.779	1.969356	0.917616	0.961986
DOR	69.025862	27.583702	431.728	1.965474	14.810814	123.240910
LR ⁺	30.780085	11.388304	1870.729	1.961233	8.444968	53.115202
LR ⁻	0.445921	0.059250	62.052	1.998938	0.327484	0.564358
J	0.543958	0.057430	58.962	2.001023	0.429040	0.658876
BA	0.771979	0.028715	58.962	2.001023	0.714520	0.829438
CZ	0.551955	0.056918	61.241	1.999464	0.438150	0.665760
FMI	0.677141	0.043363	73.825	1.992622	0.590734	0.763548
κ	0.627402	0.048943	61.350	1.999393	0.529547	0.725258
PABAK	0.858084	0.021397	253.283	1.969374	0.815945	0.900224
F ₁	0.665597	0.045887	61.689	1.999172	0.573860	0.757334
MCC	0.640998	0.046474	76.445	1.991485	0.548444	0.733551

Figure 10. Point estimates and approximate pointwise 95% expanded-uncertainty intervals for the 16 diagnostic accuracy measures at 126 mg/dL under the lognormal model.

3.10. Optimal diagnostic thresholds

CZ is maximized at 114.859 mg/dL, J and BA at 116.948 mg/dL, and ACC and PABAK at 130.011 mg/dL. Although the first-order threshold derivative vanishes at a smooth optimum, nonlinear propagation retains a measurement contribution. For ACC, the measurement and combined standard uncertainties are 0.000568 and 0.010659, respectively. The intervals are conditional on the estimated optimal threshold and exclude uncertainty from repeated threshold selection.

Measure	t^* (mg/dL)	Maximum	u_c	Lower	Upper
ACC	130.011	0.931528	0.010644	0.910557	0.952499
J	116.948	0.589408	0.054910	0.479386	0.699429
BA	116.948	0.794704	0.027455	0.739693	0.849715
CZ	114.859	0.619042	0.048186	0.522467	0.715618
FMI	128.416	0.680412	0.042300	0.595891	0.764932
κ	126.529	0.627630	0.049068	0.529471	0.725789
PABAK	130.011	0.863057	0.021288	0.821115	0.904998
F ₁	125.753	0.665641	0.045787	0.574129	0.757154
MCC	129.421	0.648410	0.044111	0.560207	0.736613

Figure 11. Optimal thresholds and corresponding measure values subject to $t > m_H$. Combined and measurement standard uncertainties and approximate 95% intervals of the measures.

4. Discussion

4.1. Principal findings

The framework propagates uncertainty for sixteen measures across nine population-family pairs. The PPV and NPV results illustrate the joint effects of mean–standard deviation covariance and a shared threshold.

Maximizing a point estimate need not minimize combined or relative uncertainty. At a smooth optimum, the first-order threshold derivative may vanish while nonlinear threshold variation still contributes to uncertainty. Neither derivative cancellation nor a high estimate establishes clinical preference.

4.2. Relations between measures

The sixteen outputs depend on Se, Sp, and prevalence, so their first-order covariance has rank at most three. They provide complementary interpretations rather than sixteen independent assessments. The identities relating BA to J and PABAK to ACC explain their common maximizing thresholds and proportional uncertainties.

4.3. Uncertainty decomposition

Retaining dependence within each source block affects the resulting variance. For PPV, accounting for covariance between the estimated mean and standard deviation decreases the variance contribution from the diseased population and increases the corresponding contribution from the nondiseased population. Threshold contribution combines both tail derivatives before squaring, accounting for the single calibration input.

Excluding prevalence uncertainty changes the combined standard uncertainty of PPV from 0.059962 to 0.055657 and that of NPV from 0.011265 to 0.007517, with unchanged central estimates. The sampling interpretation determines whether this source is included; the smaller uncertainty is not a selection criterion.

4.4. Additional sources of uncertainty and bias

Reference classification: Group labels are treated as fixed. OGTT misclassification, reference-standard uncertainty, verification bias, and incomplete disease ascertainment are excluded [28].

Distributional specification: Normal, lognormal, and gamma models may inadequately represent mixtures, truncation, censoring, or heavy tails. Checks across the nine family pairs verify executable support, not fit to participant data [11–13, 29].

Prevalence and transportability: Modeled-cohort uncertainty assumes a binomial count ratio independent of the fitted summaries. Fixed strata, survey sampling, and deployment populations may require different prevalence inputs and additional covariance terms.

Boundary behavior: First-order intervals may perform poorly near zero denominators or parameter boundaries. Transformed, likelihood-based, joint Monte Carlo, and bootstrap alternatives require separate assumptions and validation; admissible-range restriction alone is insufficient [30–33].

Sampling approximations: The sample-mean variance is exact under the stated independent-sample model with finite variance; covariance terms involving the sample standard deviation are approximated to first order. Normality does not make the propagated DAM uncertainty or interval exact.

Threshold selection: Selecting a threshold from fitted inputs introduces uncertainty and dependence omitted by a conditional interval. Resampling should repeat model estimation and threshold selection in each replicate [34–36].

Calibration transfer: Coefficients b_0 and b_1 are treated as known and shared by both populations. Propagating their estimation covariance, matrix or instrument changes, or correlations with fitted summaries requires an expanded input model.

Dependence: The model includes within-population mean–standard deviation covariance but excludes cross-population covariance, clustering, repeated measurements, and dependence between prevalence and fitted summaries. The threshold is assumed to be independent of fitted summaries.

Decision rule and intervals: Reversing the positive-test orientation requires redefining both tails. Pointwise intervals across measures and thresholds are not simultaneous intervals; admissible-range restriction does not establish nominal coverage.

4.5. Implementation strengths

Common parameterization: Arithmetic means and standard deviations retain the same interpretation across all three population families.

Source accounting: Both population blocks, optional prevalence, and the shared threshold enter each measure, except where contributions are structurally zero or explicitly disabled.

Scalar–joint agreement: The joint covariance diagonal reproduces scalar combined variances to numerical precision, and source covariance matrices sum to the total.

Reproducibility: The source and calculations Wolfram notebooks document the calculations.

4.6. Contribution

The contribution is a consistent, reproducible sixteen-measure implementation of established uncertainty methods under explicit assumptions.

4.7. Reporting recommendations

- a) Specify the positive-test orientation, population families, arithmetic summaries, sample sizes, and prevalence interpretation.
- b) Report point estimates, source-specific and combined standard uncertainties, effective degrees of freedom, coverage factors, interval levels, and any endpoint restriction.
- c) Distinguish fixed-threshold intervals from intervals conditional on a fitted optimum. Assess nonlinear and selection effects through a specified joint simulation or resampling analysis where needed.
- d) Specify the threshold standard-uncertainty model, coefficient values and units, and treatment of coefficient uncertainty and finite measurement degrees of freedom.

4.8. Limitations and future work

The calculations are based on rounded illustrative inputs; Section 3.1 and Figure 2 describe the reconstruction of the empirical NHANES distributions and the corresponding adjusted distribution fits. Interpretation is limited by first-order uncertainty propagation, effective degrees of freedom derived from source-specific variance contributions, calibration coefficients treated as known, and intervals conditional on the estimated optimal threshold. Future work should evaluate interval coverage through simulation, propagate uncertainty and covariance in the calibration coefficients, incorporate survey design and reference-standard uncertainty, and repeat model estimation and threshold selection in each bootstrap replicate. [36–42].

5. Conclusion

The framework quantifies uncertainty for sixteen diagnostic accuracy measures under independently specified normal, lognormal, or gamma population models, retaining within-population mean–standard deviation covariance and shared-threshold dependence.

Sampling contributes substantially, and threshold cancellation varies by measure; these results do not establish universal dominance across measures or thresholds.

The joint covariance represents dependence among all sixteen outputs. Checks across all nine family pairs, scalar–joint agreement, and source-matrix addition support implementation consistency.

Model-based maxima permit comparison of diagnostic objectives within the optimization domain. Clinical threshold selection additionally requires evidence on consequences, transportability, and uncertainty from estimation and selection.

6. Supplemental material

The following supplemental files are available for download as a ZIP archive at:
<https://www.hcsl.com/Supplements/SDAMUR.zip> (accessed on September 17, 2026):

- a) Supplemental File I:
DiagAccU.nb: The program as a Wolfram Mathematica Notebook.
- b) Supplemental File II:
DiagAccUCalculations.nb: The calculations for the estimation of the DAMs and their combined standard uncertainty in a Wolfram Mathematica Notebook.
- c) Supplemental File III:
DiagAccUInterface.pdf: A brief documentation of the interface of the program.

7. Declarations

Author Contributions: Conceptualization: RCA and TC; methodology: RCA and ATH; software: RCA, TC and ATH; validation: TC; formal analysis: RCA and ATH; investigation: TC; resources: ATH; data curation: TC; writing—original draft preparation: RCA; writing—review and editing: ATH; visualization: RCA and TC; supervision: ATH; project administration: RCA. All authors have read and agreed to the published version of the manuscript.

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Informed Consent Statement: Written consent was obtained from each survey participant.

Consent for publication: Not applicable.

Data Availability Statement: NHANES 2005–2016 participant data are publicly available from the National Center for Health Statistics [27]. Tables 3 and 4 provide the approximate population inputs and settings used in the illustrative calculations.

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8. References

1. Stanley DE, Campos DG. The logic of medical diagnosis. *Perspect Biol Med*. 2013;56:300–15.
2. Zou KH, O'Malley AJ, Mauri L. Receiver-operating characteristic analysis for evaluating diagnostic tests and predictive models. *Circulation*. 2007;115:654–7.

3. Djulbegovic B, van den Ende J, Hamm RM, Mayrhofer T, Hozo I, Pauker SG, et al. When is rational to order a diagnostic test, or prescribe treatment: the threshold model as an explanation of practice variation. *Eur J Clin Invest*. 2015;45:485–93.
4. Šimundić A-M. Measures of Diagnostic Accuracy: Basic Definitions. *EJIFCC*. 2009;19:203–11.
5. Larner AJ. The 2x2 matrix: Contingency, confusion and the metrics of binary classification. Cham: Springer International Publishing; 2024.
6. BIPM, IEC, IFCC, ILAC, ISO, IUPAC, et al. Evaluation of measurement data --- Guide to the expression of uncertainty in measurement. 2008. <https://doi.org/10.59161/JCGM100-2008E>.
7. Kallner A, Boyd JC, Duewer DL, Giroud C, Hatjimihail AT, Klee GG. Expression of Measurement Uncertainty in Laboratory Medicine. 2012;:1–56.
8. Faraggi D. The effect of random measurement error on receiver operating characteristic (ROC) curves. *Stat Med*. 2000;19:61–70.
9. Reiser B. Measuring the effectiveness of diagnostic markers in the presence of measurement error through the use of ROC curves. *Stat Med*. 2000;19:2115–29.
10. Perkins NJ, Schisterman EF. The Youden Index and the optimal cut-point corrected for measurement error. *Biom J*. 2005;47:428–41.
11. Schmoyer RL, Beauchamp JJ, Brandt CC, Hoffman FO Jr. Difficulties with the lognormal model in mean estimation and testing. *Environ Ecol Stat*. 1996;3:81–97.
12. Bhaumik DK, Kapur K, Gibbons RD. Testing Parameters of a Gamma Distribution for Small Samples. *Technometrics*. 2009;51:326–34.
13. Goddard MJ, Hinberg I. Receiver operator characteristic (ROC) curves and non-normal data: an empirical study. *Stat Med*. 1990;9:325–37.
14. Theodorsson E. Uncertainty in Measurement and Total Error: Tools for Coping with Diagnostic Uncertainty. *Clin Lab Med*. 2017;37:15–34.
15. Padoan A, Sciacovelli L, Aita A, Antonelli G, Plebani M. Measurement uncertainty in laboratory reports: A tool for improving the interpretation of test results. *Clin Biochem*. 2018;57:41–7.
16. Smith AF, Shinkins B, Hall PS, Hulme CT, Messenger MP. Toward a Framework for Outcome-Based Analytical Performance Specifications: A Methodology Review of Indirect Methods for Evaluating the Impact of Measurement Uncertainty on Clinical Outcomes. *Clin Chem*. 2019;65:1363–74.
17. Chatzimichail T, Hatjimihail AT. A Software Tool for Calculating the Uncertainty of Diagnostic Accuracy Measures. *Diagnostics (Basel)*. 2021;11. <https://doi.org/10.3390/diagnostics11030406>.
18. Chatzimichail T, Hatjimihail AT. A Software Tool for Exploring the Relation between Diagnostic Accuracy and Measurement Uncertainty. *Diagnostics (Basel)*. 2020;10. <https://doi.org/10.3390/diagnostics10090610>.
19. Chatzimichail T, Hatjimihail AT. A software tool for applying Bayes' theorem in medical diagnostics. *BMC Med Inform Decis Mak*. 2024;24:399.
20. Welch BL. The Generalization of 'Student's' Problem when Several Different Population Variances are Involved. *Biometrika*. 1947;34:28–35.
21. Satterthwaite FE. An approximate distribution of estimates of variance components. *Biometrics*. 1946;2:110–4.
22. Agresti A, Franklin C, Klingenberg B. Statistics: The art and science of learning from data, global edition. 4th edition. London, England: Pearson Education; 2023.
23. Miller J, Miller JC. Statistics and Chemometrics for Analytical Chemistry. 7th edition. London, England: Pearson Education; 2018.
24. Aitchison J, Brown JAC. The Lognormal Distribution with special reference to its uses in econometrics. Cambridge: Cambridge University Press; 1957.
25. AroundReplace. Wolfram Language & System Documentation Center. <https://reference.wolfram.com/>. Accessed 14 Sept 2026.
26. VectorAround. Wolfram Language & System Documentation Center. <https://reference.wolfram.com/>. Accessed 14 Sept 2026.
27. National Health and Nutrition Examination Survey Data. Centers for Disease Control and Prevention, National Center for Health Statistics. 2005-2016. <https://wwwn.cdc.gov/nchs/nhanes/default.aspx>. Accessed 4 June 2026.

28. Rao SS, Disraeli P, McGregor T. Impaired glucose tolerance and impaired fasting glucose. *Am Fam Physician*. 2004;69:1961–8.
29. Berlin KS, Williams NA, Parra GR. An introduction to latent variable mixture modeling (part 1): overview and cross-sectional latent class and latent profile analyses. *J Pediatr Psychol*. 2014;39:174–87.
30. West RM. Best practice in statistics: The use of log transformation. *Ann Clin Biochem*. 2022;59:162–5.
31. Fieller EC. The distribution of the index in a normal bivariate population. *Biometrika*. 1932;24:428–40.
32. Pawitan Y. In all likelihood: Statistical modelling and inference using likelihood. Oxford University PressOxford; 2001.
33. Dikta G, Scheer M. Bootstrap methods: With applications in R. 2021st edition. Cham, Switzerland: Springer Nature; 2021.
34. Ambrose C, McLachlan GJ. Selection bias in gene extraction on the basis of microarray gene-expression data. *Proc Natl Acad Sci U S A*. 2002;99:6562–6.
35. Janes H, Pepe MS, Bossuyt PM, Barlow WE. Measuring the performance of markers for guiding treatment decisions. *Ann Intern Med*. 2011;154:253–9.
36. Davison AC, Hinkley DV. Cambridge series in statistical and probabilistic mathematics: Bootstrap methods and their application series number 1. Cambridge, England: Cambridge University Press; 2013. <https://doi.org/10.1017/cbo9780511802843>.
37. Branscum AJ, Johnson WO, Hanson TE, Gardner IA. Bayesian semiparametric ROC curve estimation and disease diagnosis. *Stat Med*. 2008;27:2474–96.
38. Zheng J, Frey HC. Quantification of variability and uncertainty using mixture distributions: evaluation of sample size, mixing weights, and separation between components. *Risk Anal*. 2004;24:553–71.
39. Hand DJ, Till RJ. A simple generalisation of the area under the ROC curve for multiple class classification problems. *Mach Learn*. 2001;45:171–86.
40. Thiele C, Hirschfeld G. cutpointr: Improved Estimation and Validation of Optimal Cutpoints in R. *J Stat Softw*. 2021;98:1–27.
41. Vickers AJ, Elkin EB. Decision curve analysis: a novel method for evaluating prediction models. *Med Decis Making*. 2006;26:565–74.
42. Vickers AJ, van Calster B, Steyerberg EW. A simple, step-by-step guide to interpreting decision curve analysis. *Diagn Progn Res*. 2019;3:18.
43. Chatzimichail RA, Chatzimichail T, Hatjimihail AT. DiagAccU: a unified framework for uncertainty estimation of diagnostic accuracy measures. Version 2.0.0. Hellenic Complex Systems Laboratory; 2026. doi: 10.5281/zenodo.22817814.
44. Joint Committee for Guides in Metrology. Evaluation of measurement data: Supplement 2 to the Guide to the expression of uncertainty in measurement—Extension to any number of output quantities. JCGM 102:2011. 2011. doi:10.59161/JCGM102-2011.
45. Pewsey A. The large-sample distribution of the most fundamental of statistical summaries. *J Stat Plan Inference*. 2005;134(2):434–44.

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Appendix A. Mathematical and statistical calculations

This appendix presents the mathematical and statistical calculations from input specification to expanded-uncertainty intervals. Section A.5 provides the complete derivative table, and Section A.7 gives the joint covariance formulation.

Variance is the square of the corresponding standard uncertainty. Covariance describes how two uncertain quantities vary together. A partial derivative describes the local rate of change of an output with respect to one input, with all other inputs held fixed. A gradient collects the partial derivatives for one output, and a Jacobian collects them for several outputs. First-order uncertainty propagation is based on a local linear approximation.

A.1. Input specification and ordering

The population labels D and H denote diseased and nondiseased populations, respectively. Prevalence is denoted v , and the intermediate arguments a and b denote Se and Sp. A generic diagnostic accuracy measure is denoted F . The standard uncertainties u_c , u_s , and u_m , evaluated for a specified measure, denote combined, total sampling, and measurement uncertainty, respectively. Measurement uncertainty is modeled through a diagnostic threshold shared by both populations. The population-parameter and prevalence sampling components are denoted u_{par} and u_{prev} . Superscript T denotes transpose. The measure is a function of the primitive inputs, $F = g(\mathbf{x})$.

Scalar variables are italic; measure abbreviations, named operators, and descriptive subscripts are upright. Vectors and matrices are bold. The primitive-input Jacobian is denoted $\mathbf{M}$, and the four source-specific variance contributions are denoted V_D , V_H , V_v , and V_t . These correspond to the Jacobian $\mathbf{J}$ and the variance contributions v_D , v_H , v_v , and v_t , respectively, in Appendix B. The symbol J denotes Youden's index throughout this appendix.

The primitive input vector comprises the four fitted arithmetic summaries, prevalence, and the common diagnostic threshold. The intermediate vector comprises the three arguments of each diagnostic accuracy measure.

$$\begin{aligned}\mathbf{x} &= \begin{pmatrix} m_D \\ s_D \\ m_H \\ s_H \\ v \\ t \end{pmatrix} \in \mathbb{R}^6, \\ \mathbf{z} &= \begin{pmatrix} a \\ b \\ v \end{pmatrix} = \begin{pmatrix} \text{Se} \\ \text{Sp} \\ v \end{pmatrix} \in \mathbb{R}^3.\end{aligned}\tag{A.1}$$

The primitive and intermediate vectors have six and three components, respectively. All vectors are column vectors. The aliases $a = \text{Se}$ and $b = \text{Sp}$ are used in the derivative formulas; v denotes prevalence throughout.

$$\mathbf{m} = \begin{pmatrix} \text{Se} \\ \text{Sp} \\ \text{ACC} \\ \text{PPV} \\ \text{NPV} \\ \text{DOR} \\ \text{LR}^+ \\ \text{LR}^- \\ J \\ \text{BA} \\ \text{CZ} \\ \text{FMI} \\ \kappa \\ \text{PABAK} \\ F_1 \\ \text{MCC} \end{pmatrix} \in \mathbb{R}^{16}. \quad (\text{A.2})$$

Equation (A.2) specifies the output order. Row i of each output Jacobian corresponds to measure i , and element (i, j) of the output covariance matrix is the covariance between measures i and j .

Observations are assumed to be independent within each population, and the population samples are assumed to be independent of one another. The supplied standard deviations are assumed to be calculated from Bessel-corrected sample variances. Prevalence and the diagnostic threshold are modeled as mutually independent inputs, each independent of the fitted population summaries. These assumptions specify which elements of the input covariance matrix are zero. For other study designs, the covariance matrix must reflect the corresponding dependence structure.

A.2. Distribution fitting and calculation of the two classification rates

For population $P \in \{D, H\}$, the normal, lognormal, and gamma families are parameterized by the arithmetic mean μ_P and standard deviation σ_P . Lognormal and gamma models require positive means, and positive thresholds are required whenever either population has positive support. The gamma parameters ξ and θ denote shape and scale, respectively.

$$\begin{aligned} X_N &\sim \mathcal{N}(\mu, \sigma^2). \\ \sigma_L^2 &= \log \left(1 + \left(\frac{\sigma}{\mu} \right)^2 \right), \\ \mu_L &= \log(\mu) - \frac{\sigma_L^2}{2}. \\ X_L &\sim \text{Lognormal}(\mu_L, \sigma_L). \\ \xi &= \left(\frac{\mu}{\sigma} \right)^2, \quad \theta = \frac{\sigma^2}{\mu}, \\ X_G &\sim \text{Gamma}(\xi, \theta). \end{aligned}$$

The selected distribution family is parameterized using the supplied arithmetic mean and standard deviation. For the lognormal family, the transformed parameters describe the logarithm of the measurement. For the gamma family, the shape and scale are obtained from the two arithmetic summaries. Agreement with the arithmetic summaries does not establish adequate fit in the distribution tails.

For a positive-above-threshold test, Se is the probability in the diseased upper tail and Sp is the probability in the nondiseased lower tail. A cumulative distribution function gives the probability below a threshold. Thus Se is one minus the diseased cumulative probability, whereas Sp is the nondiseased cumulative probability.

The class-conditional cumulative distribution functions are denoted F_D and F_H . Under the positive-above-threshold rule, Se and Sp are defined as follows:

$$\text{Se}(t) = \Pr(X_D \geq t).$$

$$\text{Sp}(t) = \Pr(X_H < t).$$

$$\text{Se}(t) = 1 - F_D(t), \quad \text{Sp}(t) = F_H(t).$$

For continuous distributions, including all three families used here, the probability of observing the threshold exactly is zero. The distinction between a strict and a non-strict inequality therefore does not alter the model probabilities.

A.3. Calculation of diagnostic accuracy measures

Tables A1 and A2 define the remaining measures. The symbols v , q , p_o , and p_e denote prevalence, predicted-positive probability, observed agreement, and chance agreement, respectively:

$$q = v \text{Se} + (1 - v) (1 - \text{Sp}).$$

$$p_o = v \text{Se} + (1 - v) \text{Sp}.$$

$$p_e = q v + (1 - q) (1 - v).$$

Table A1. Mathematical definitions of Se, Sp, PPV, NPV, ACC, LR⁺, LR[−], and balance measures.

Measure	Count definition	Definition using Se, Sp, v
Se	$\frac{\text{TP}}{\text{TP} + \text{FN}}$	Se
Sp	$\frac{\text{TN}}{\text{TN} + \text{FP}}$	Sp
PPV	$\frac{\text{TP}}{\text{TP} + \text{FP}}$	$\frac{v\text{Se}}{q}$
NPV	$\frac{\text{TN}}{\text{TN} + \text{FN}}$	$\frac{(1 - v)\text{Sp}}{1 - q}$
ACC	$\frac{\text{TP} + \text{TN}}{N}$	$v\text{Se} + (1 - v)\text{Sp}$
DOR	$\frac{\text{TPTN}}{\text{FPFN}}$	$\frac{\text{Se Sp}}{(1 - \text{Se})(1 - \text{Sp})}$
LR ⁺	$\frac{\frac{\text{TP}}{R}}{\frac{\text{FP}}{N - R}}$	$\frac{\text{Se}}{1 - \text{Sp}}$
LR [−]	$\frac{\frac{\text{FN}}{R}}{\frac{\text{TN}}{N - R}}$	$\frac{1 - \text{Se}}{\text{Sp}}$
J	$\frac{\text{TP}}{R} + \frac{\text{TN}}{N - R} - 1$	$\text{Se} + \text{Sp} - 1$
BA	$\frac{\frac{\text{TP}}{R} + \frac{\text{TN}}{N - R}}{2}$	$\frac{\text{Se} + \text{Sp}}{2}$
CZ	$\frac{\text{TPTN}}{R(N - R)}$	Se Sp

Table A2. Mathematical definitions of association and agreement measures.

Measure	Count definition	Definition using Se, Sp, v
FMI	$\frac{TP}{\sqrt{RN_+}}$	$\sqrt{Se PPV}$
κ	$\frac{N(TP + TN) - RN_+ - (N - R)(N - N_+)}{N^2 - RN_+ - (N - R)(N - N_+)}$	$\frac{p_o - p_e}{1 - p_e}$
PABAK	$2 \frac{TP + TN}{N} - 1$	$2p_o - 1$
F_1	$\frac{2TP}{2TP + FP + FN}$	$\frac{2vSe}{v + q}$
MCC	$\frac{TPTN - FPFN}{\sqrt{RN_+(N - R)(N - N_+)}}$	$\frac{\sqrt{v(1 - v)}(Se + Sp - 1)}{\sqrt{q(1 - q)}}$

In Tables A1 and A2, $N = TP + FN + FP + TN$, $R = TP + FN$, and $N_+ = TP + FP$ denote total, disease-positive marginal, and test-positive marginal counts, respectively. Dividing counts by N gives the equivalent probability definitions. Ratios and their derivatives require nonzero denominators.

Classification of measures

Table A3. Overlapping classification of the sixteen measures.

Category	Measures
Agreement	κ , PABAK
Association and F-type indices	FMI, F_1 , MCC
Balance	J , BA, CZ
Bounded	Se, Sp, ACC, PPV, NPV, J , BA, CZ, FMI, κ , PABAK, F_1 , MCC
Ratio-type	DOR, LR^+ , LR^-
Prevalence invariant	Se, Sp, DOR, LR^+ , LR^- , J , BA, CZ
Prevalence dependent	ACC, PPV, NPV, FMI, κ , PABAK, F_1 , MCC

The classifications overlap: agreement measures and F-type indices may depend on prevalence. Boundedness implies neither low uncertainty nor robustness to class imbalance.

The predicted-positive probability equals the sum of the probabilities of true-positive and false-positive outcomes in the target population. PPV is the ratio of the true-positive probability to this sum. Similarly, NPV is the ratio of the true-negative probability to the sum of the true-negative and false-negative probabilities. Agreement and association measures are also calculated from these outcome probabilities.

A.4. Construction of the input covariance matrix

The input covariance matrix includes the variances of the estimated means and standard deviations for both populations, together with the prevalence and threshold variances. Within each population, the covariance between the estimated mean and standard deviation is retained. All remaining covariances are zero under the stated independence assumptions [6, 44, 45].

A.4.1. Sampling uncertainty within each population

$$\begin{aligned}\Sigma_{\bar{X},S} &= E \left(\begin{pmatrix} \delta_{\bar{X}} \\ \delta_S \end{pmatrix} \begin{pmatrix} \delta_{\bar{X}} & \delta_S \end{pmatrix} \right), \\ \Sigma_{\bar{X},S} &= E \left(\begin{pmatrix} \delta_{\bar{X}}^2 & \delta_{\bar{X}} \delta_S \\ \delta_S \delta_{\bar{X}} & \delta_S^2 \end{pmatrix} \right), \\ \Sigma_{\bar{X},S} &\approx \begin{pmatrix} A & B \\ B & C \end{pmatrix}.\end{aligned}\tag{A.3}$$

The covariance matrix is the expectation of the outer product of the centered estimator vector with its transpose. Expectations are evaluated for each matrix element. The off-diagonal elements are equal by symmetry of covariance.

$$\begin{aligned}\Sigma_{\bar{X},S^2} &= \begin{pmatrix} \frac{\sigma^2}{n} & \frac{\mu_3}{n} \\ \frac{\mu_3}{n} & \frac{\mu_4 - \frac{n-3}{n-1} \sigma^4}{n} \end{pmatrix}, \\ \mathbf{L} &= \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2\sigma} \end{pmatrix}.\end{aligned}\tag{A.4}$$

The transformation starts from the covariance of the sample mean and Bessel-corrected sample variance. Its Jacobian $\mathbf{L}$ leaves the mean unchanged and maps variance to its positive square root.

$$\mathbf{L} \Sigma_{\bar{X},S^2} = \begin{pmatrix} \frac{\sigma^2}{n} & \frac{\mu_3}{n} \\ \frac{\mu_3}{2n\sigma} & \frac{\mu_4 - \frac{n-3}{n-1} \sigma^4}{2n\sigma} \end{pmatrix}.\tag{A.5}$$

Left multiplication by $\mathbf{L}$ divides the second row by twice the population standard deviation.

$$\begin{aligned}\Sigma_{\bar{X},S} &\approx \mathbf{L} \Sigma_{\bar{X},S^2} \mathbf{L}^T, \\ \Sigma_{\bar{X},S} &\approx \begin{pmatrix} \frac{\sigma^2}{n} & \frac{\mu_3}{2n\sigma} \\ \frac{\mu_3}{2n\sigma} & \frac{\mu_4 - \frac{n-3}{n-1} \sigma^4}{4n\sigma^2} \end{pmatrix}.\end{aligned}\tag{A.6}$$

Right multiplication by the transpose of $\mathbf{L}$ scales the second column. Together, the two multiplications divide the mean–variance covariance by twice the population standard deviation and the variance of the sample variance by four times the population variance, yielding the first-order mean–standard deviation covariance matrix.

$$\begin{aligned}A &= \frac{\sigma^2}{n}, \\ B &= \frac{\mu_3}{2n\sigma}, \\ C &= \frac{\mu_4 - \frac{n-3}{n-1} \sigma^4}{4n\sigma^2}.\end{aligned}\tag{A.7}$$

A is the exact variance of the sample mean. B and C are first-order approximations to the mean–standard deviation covariance and the variance of the sample standard deviation, respectively. Their dependence on the third and fourth central moments makes B and C distribution dependent. For a normal population, B is zero; it is generally nonzero for the lognormal and gamma families [45].

The third and fourth central moments reflect distributional asymmetry and the contribution of large deviations from the mean, respectively. The family-specific expressions below are evaluated using the estimated arithmetic means and standard deviations. The reported uncertainty is therefore obtained by substituting parameter estimates into these expressions, even when the underlying variance identities are exact. For the lognormal expressions, $c = 1 + \left(\frac{\sigma}{\mu}\right)^2$.

$$\begin{aligned}
&\text{Normal: } \mu_3 = 0, \mu_4 = 3\sigma^4; \\
&\text{Lognormal: } \mu_3 = \frac{(c+2)\sigma^4}{\mu}, \mu_4 = (c^4 + 2c^3 + 3c^2 - 3)\sigma^4; \\
&\text{Gamma: } \mu_3 = \frac{2\sigma^4}{\mu}, \mu_4 = 3\sigma^4 + \frac{6\sigma^6}{\mu^2}.
\end{aligned}$$

$$\begin{aligned}
\mathbf{\Sigma}_D &= \begin{pmatrix} A_D & B_D \\ B_D & C_D \end{pmatrix}, \\
\mathbf{\Sigma}_H &= \begin{pmatrix} A_H & B_H \\ B_H & C_H \end{pmatrix}.
\end{aligned} \tag{A.8}$$

For each population, the sample arithmetic mean precedes the sample standard deviation in the input vector. The corresponding covariance matrices are symmetric.

$$\mathbf{\Sigma}_{\text{par}} = \begin{pmatrix} A_D & B_D & 0 & 0 \\ B_D & C_D & 0 & 0 \\ 0 & 0 & A_H & B_H \\ 0 & 0 & B_H & C_H \end{pmatrix}. \tag{A.9}$$

The matrix $\mathbf{\Sigma}_{\text{par}}$ contains the variances and covariances of the four fitted population summaries. The upper-left and lower-right submatrices correspond to the diseased and nondiseased populations, respectively. Covariances between estimators from different populations are zero under the assumption of independent samples.

$$\mathbf{\Sigma}_x = \begin{pmatrix} A_D & B_D & 0 & 0 & 0 & 0 \\ B_D & C_D & 0 & 0 & 0 & 0 \\ 0 & 0 & A_H & B_H & 0 & 0 \\ 0 & 0 & B_H & C_H & 0 & 0 \\ 0 & 0 & 0 & 0 & u_v^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & u_t^2 \end{pmatrix}. \tag{A.10}$$

Including prevalence and the diagnostic threshold alongside the four fitted population summaries extends the input covariance matrix from 4×4 to 6×6 . Their variances occupy matrix elements (5,5) and (6,6), respectively. All additional off-diagonal elements are zero under the assumed independence of these inputs from the fitted population summaries and from one another. When prevalence uncertainty is excluded, element (5,5) is set to zero; the matrix dimensions and input ordering remain unchanged.

A.4.2. Prevalence uncertainty

Under the assumed cohort sampling model, prevalence and its standard uncertainty are estimated as follows:

$$\begin{aligned}
v &= \frac{n_D}{n_D + n_H}, \\
u_v &= \sqrt{\frac{v(1-v)}{n_D + n_H}}.
\end{aligned}$$

Prevalence uncertainty is included by default. For deliberately fixed case-control strata, the option `IncludePrevalenceUncertainty → False` removes only this source. The central count ratio is unchanged; PPV and NPV for another setting require a prevalence appropriate to that target population.

A.4.3. Measurement uncertainty

The threshold standard uncertainty is specified by either a linear model, restricted to nonnegative values, or a root-sum-of-squares model. The coefficient b_0 is the constant component and has the measurement unit; b_1 is the dimensionless proportional coefficient. Both coefficients are treated as known. Gauss–Hermite quadrature is a separate procedure for propagating measurement uncertainty at an estimated optimal threshold.

$$\begin{aligned}
 u_t &= b_0 + b_1 t \\
 &\text{or} \\
 u_t &= \sqrt{b_0^2 + (b_1 t)^2}.
 \end{aligned}$$

$$\mathbf{h}_t = \begin{pmatrix} \frac{\partial \text{Se}}{\partial t} \\ \frac{\partial \text{Sp}}{\partial t} \end{pmatrix} = \begin{pmatrix} -f_D(t) \\ f_H(t) \end{pmatrix}. \quad (\text{A.11})$$

The threshold derivative of Se is nonpositive, whereas that of Sp is nonnegative. The 2×1 sensitivity vector and its transpose determine the covariance contribution from the common threshold.

$$\begin{aligned}
 \Sigma_{\text{Se,Sp};t} &= u_t^2 \mathbf{h}_t \mathbf{h}_t^T, \\
 \Sigma_{\text{Se,Sp};t} &= u_t^2 \begin{pmatrix} -f_D(t) \\ f_H(t) \end{pmatrix} \begin{pmatrix} -f_D(t) & f_H(t) \end{pmatrix}.
 \end{aligned} \quad (\text{A.12})$$

$$\begin{aligned}
 \Sigma_{\text{Se,Sp};t} &= u_t^2 \begin{pmatrix} (-f_D(t))(-f_D(t)) & (-f_D(t))f_H(t) \\ f_H(t)(-f_D(t)) & f_H(t)f_H(t) \end{pmatrix}, \\
 \Sigma_{\text{Se,Sp};t} &= u_t^2 \begin{pmatrix} f_D(t)^2 & -f_D(t)f_H(t) \\ -f_D(t)f_H(t) & f_H(t)^2 \end{pmatrix}.
 \end{aligned} \quad (\text{A.13})$$

The diagonal elements are the squared population densities multiplied by the threshold variance. The off-diagonal elements are the negative product of the two densities multiplied by the same variance. The resulting covariance matrix has rank one whenever the threshold variance is positive and at least one density is nonzero.

The covariance induced by the common threshold is nonpositive: an increase in the threshold decreases Se and increases Sp wherever the corresponding densities are positive. The threshold variance and both marginal variance contributions remain nonnegative.

A.5. Calculation of the derivatives

The partial derivatives of each measure with respect to the six primitive inputs are obtained from its derivatives with respect to the three intermediate inputs by the chain rule.

$$\begin{aligned}
 a &= \text{Se}, & b &= \text{Sp}, \\
 J &= a + b - 1, \\
 q &= v a + (1 - v)(1 - b), \\
 p_o &= v a + (1 - v)b, \\
 p_e &= (1 - v) + (2v - 1)q, \\
 \eta &= v + q.
 \end{aligned} \quad (\text{A.14})$$

$$\begin{aligned}
\text{PPV} &= \frac{v a}{q}, & \text{NPV} &= \frac{(1-v) b}{(1-q)}, \\
\text{FMI} &= a \sqrt{\frac{v}{q}}, \\
r_v &= \sqrt{v(1-v)}, & r_q &= \sqrt{q(1-q)}, \\
\text{MCC} &= \frac{r_v J}{r_q}.
\end{aligned} \tag{A.15}$$

The quantities q , p_o , and p_e denote predicted-positive probability, observed agreement, and chance agreement, respectively. The denominator in F_1 is denoted $\eta = v + q$; its square appears in the derivative denominators. For MCC, r_v and r_q denote the square-root factors associated with prevalence and predicted-positive probability, respectively; J retains its definition as Youden's index. PPV, NPV, and FMI retain their established definitions.

$$\begin{aligned}
q_a &= v, & q_b &= -(1-v), & q_v &= J; \\
p_{o_a} &= v, & p_{o_b} &= (1-v), & p_{o_v} &= a-b; \\
p_{e_a} &= v(2v-1), \\
p_{e_b} &= (1-v)(1-2v), \\
p_{e_v} &= -1 + 2q + (2v-1)J.
\end{aligned} \tag{A.16}$$

The derivatives of the three probabilities follow from their definitions. The derivative of predicted-positive probability with respect to Sp is nonpositive. The prevalence derivative of chance agreement includes both contributions from the product rule.

Table A4. All 48 elements of the 16×3 diagnostic-measure Jacobian

Measure	Jacobian element	Expanded scalar derivative
Se	$\frac{\partial \text{Se}}{\partial a}$	1
Se	$\frac{\partial \text{Se}}{\partial b}$	0
Se	$\frac{\partial \text{Se}}{\partial v}$	0
Sp	$\frac{\partial \text{Sp}}{\partial a}$	0
Sp	$\frac{\partial \text{Sp}}{\partial b}$	1
Sp	$\frac{\partial \text{Sp}}{\partial v}$	0
ACC	$\frac{\partial \text{ACC}}{\partial a}$	v
ACC	$\frac{\partial \text{ACC}}{\partial b}$	$(1-v)$
ACC	$\frac{\partial \text{ACC}}{\partial v}$	$a-b$
PPV	$\frac{\partial \text{PPV}}{\partial a}$	$\frac{v(1-v)(1-b)}{q^2}$
PPV	$\frac{\partial \text{PPV}}{\partial b}$	$\frac{v a(1-v)}{q^2}$

Measure	Jacobian element	Expanded scalar derivative
PPV	$\frac{\partial \text{PPV}}{\partial v}$	$\frac{a(1-b)}{q^2}$
NPV	$\frac{\partial \text{NPV}}{\partial a}$	$\frac{v(1-v)b}{(1-q)^2}$
NPV	$\frac{\partial \text{NPV}}{\partial b}$	$\frac{v(1-v)(1-a)}{(1-q)^2}$
NPV	$\frac{\partial \text{NPV}}{\partial v}$	$-\frac{b(1-a)}{(1-q)^2}$
DOR	$\frac{\partial \text{DOR}}{\partial a}$	$\frac{b}{(1-b)(1-a)^2}$
DOR	$\frac{\partial \text{DOR}}{\partial b}$	$\frac{a}{(1-a)(1-b)^2}$
DOR	$\frac{\partial \text{DOR}}{\partial v}$	0
LR ⁺	$\frac{\partial \text{LR}^+}{\partial a}$	$\frac{1}{(1-b)}$
LR ⁺	$\frac{\partial \text{LR}^+}{\partial b}$	$\frac{a}{(1-b)^2}$
LR ⁺	$\frac{\partial \text{LR}^+}{\partial v}$	0
LR ⁻	$\frac{\partial \text{LR}^-}{\partial a}$	$-\frac{1}{b}$
LR ⁻	$\frac{\partial \text{LR}^-}{\partial b}$	$-\frac{(1-a)}{b^2}$
LR ⁻	$\frac{\partial \text{LR}^-}{\partial v}$	0
J	$\frac{\partial J}{\partial a}$	1
J	$\frac{\partial J}{\partial b}$	1
J	$\frac{\partial J}{\partial v}$	0
BA	$\frac{\partial \text{BA}}{\partial a}$	$\frac{1}{2}$
BA	$\frac{\partial \text{BA}}{\partial b}$	$\frac{1}{2}$
BA	$\frac{\partial \text{BA}}{\partial v}$	0
CZ	$\frac{\partial \text{CZ}}{\partial a}$	b
CZ	$\frac{\partial \text{CZ}}{\partial b}$	a
CZ	$\frac{\partial \text{CZ}}{\partial v}$	0

Measure	Jacobian element	Expanded scalar derivative
FMI	$\frac{\partial \text{FMI}}{\partial a}$	$\text{FMI} \left(\frac{1}{a} - \frac{v}{2q} \right)$
FMI	$\frac{\partial \text{FMI}}{\partial b}$	$\frac{\text{FMI} (1 - v)}{2q}$
FMI	$\frac{\partial \text{FMI}}{\partial v}$	$\text{FMI} \left(\frac{1}{2v} - \frac{J}{2q} \right)$
κ	$\frac{\partial \kappa}{\partial a}$	$\frac{(1 - p_e) v + (p_o - 1) v (2v - 1)}{(1 - p_e)^2}$
κ	$\frac{\partial \kappa}{\partial b}$	$\frac{(1 - p_e) (1 - v) + (p_o - 1) (1 - v) (1 - 2v)}{(1 - p_e)^2}$
κ	$\frac{\partial \kappa}{\partial v}$	$\frac{(1 - p_e) (a - b) + (p_o - 1) (-1 + 2q + (2v - 1) J)}{(1 - p_e)^2}$
PABAK	$\frac{\partial \text{PABAK}}{\partial a}$	$2v$
PABAK	$\frac{\partial \text{PABAK}}{\partial b}$	$2(1 - v)$
PABAK	$\frac{\partial \text{PABAK}}{\partial v}$	$2(a - b)$
F_1	$\frac{\partial F_1}{\partial a}$	$\frac{2v(v + (1 - v)(1 - b))}{\eta^2}$
F_1	$\frac{\partial F_1}{\partial b}$	$\frac{2va(1 - v)}{\eta^2}$
F_1	$\frac{\partial F_1}{\partial v}$	$\frac{2a(1 - b)}{\eta^2}$
MCC	$\frac{\partial \text{MCC}}{\partial a}$	$\frac{r_v}{r_q} \left(1 - \frac{J(1 - 2q)v}{2q(1 - q)} \right)$
MCC	$\frac{\partial \text{MCC}}{\partial b}$	$\frac{r_v}{r_q} \left(1 + \frac{J(1 - 2q)(1 - v)}{2q(1 - q)} \right)$
MCC	$\frac{\partial \text{MCC}}{\partial v}$	$\text{MCC} \left(\frac{(1 - 2v)}{2v(1 - v)} - \frac{J(1 - 2q)}{2q(1 - q)} \right)$

Rows follow the output order in equation (A.2). The three derivatives for each measure are taken with respect to a , b , and v . Denominators must be nonzero; expressions containing reciprocals or square roots require parameter values in their differentiability domains.

$$\mathbf{G} = \frac{\partial \mathbf{m}}{\partial \mathbf{z}^T} \in \mathbb{R}^{16 \times 3}, \quad (\text{A.17})$$

$$\mathbf{G}_{i \cdot} = \left(\frac{\partial F_i}{\partial a} \quad \frac{\partial F_i}{\partial b} \quad \frac{\partial F_i}{\partial v} \right) = (G_{ia} \quad G_{ib} \quad G_{iv}).$$

Table A4 supplies all elements of $\mathbf{G}$. The 16×1 output vector is differentiated with respect to the transpose of the 3×1 intermediate vector $\mathbf{z}$, yielding a 16×3 Jacobian.

A.5.1. Chain-rule expansion from three intermediate to six primitive inputs

$$\begin{aligned} a_m &= \frac{\partial a}{\partial m_D}, & a_s &= \frac{\partial a}{\partial s_D}, \\ b_m &= \frac{\partial b}{\partial m_H}, & b_s &= \frac{\partial b}{\partial s_H}. \end{aligned} \quad (\text{A.18})$$

$$\mathbf{K} = \frac{\partial \mathbf{z}}{\partial \mathbf{x}^T} = \begin{pmatrix} a_m & a_s & 0 & 0 & 0 & -f_D(t) \\ 0 & 0 & b_m & b_s & 0 & f_H(t) \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}. \quad (\text{A.19})$$

The first row contains diseased-summary and threshold derivatives; the second contains nondiseased-summary and threshold derivatives. The third row selects prevalence. Population-family tail formulas supply the mean and standard-deviation derivatives. The threshold derivatives are minus the diseased density and plus the nondiseased density.

$$\begin{aligned} \mathbf{M} &= \mathbf{G} \mathbf{K} \in \mathbb{R}^{16 \times 6}, \\ \mathbf{M}_{i.} &= (G_{ia} a_m \quad G_{ia} a_s \quad G_{ib} b_m \quad G_{ib} b_s \quad G_{iv} \quad -G_{ia} f_D(t) + G_{ib} f_H(t)). \end{aligned} \quad (\text{A.20})$$

Multiplication of $\mathbf{G}$ by $\mathbf{K}$ applies the chain rule to each measure. The sixth column of $\mathbf{M}$ combines the contributions of Se and Sp to the derivative with respect to the common threshold.

Population-summary derivatives are obtained by differentiating the fitted tail probabilities with respect to the arithmetic mean or standard deviation, including the induced changes in the distribution parameters. For the lognormal family, a change in the arithmetic standard deviation changes both lognormal parameters. Threshold derivatives are obtained directly from the population densities evaluated at the threshold.

A numerical approximation to a smooth scalar derivative is obtained by a central difference, with all other inputs held fixed:

$$\frac{g(x + h \mathbf{e}_j) - g(x - h \mathbf{e}_j)}{2h} \approx \partial g / \partial x_j.$$

The unit vector $\mathbf{e}_j$ identifies the perturbed input, and the positive step h has the same unit as that input. Both perturbed inputs must remain in the admissible domain. Stability under a reduction in step size provides a numerical check, although excessively small steps may amplify rounding effects.

A.6. Uncertainty propagation for a single measure

First-order propagation combines input variances and covariances with the corresponding partial derivatives to obtain the output variance. The combined standard uncertainty is its nonnegative square root. Variance contributions from independent sources are additive; standard uncertainties are not. The scalar expansion below expresses the calculation explicitly.

$$\begin{aligned} \mathbf{d} &= \nabla g = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \\ d_6 \end{pmatrix}, \\ d_1 &= \frac{\partial F}{\partial m_D}, & d_2 &= \frac{\partial F}{\partial s_D}, \\ d_3 &= \frac{\partial F}{\partial m_H}, & d_4 &= \frac{\partial F}{\partial s_H}, \\ d_5 &= \frac{\partial F}{\partial v}, & d_6 &= \frac{\partial F}{\partial t}. \end{aligned} \quad (\text{A.21})$$

The gradient contains six partial derivatives. The propagated variance is the quadratic form obtained by multiplying the input covariance matrix by the gradient and premultiplying the result by its transpose.

$$\mathbf{w} = \boldsymbol{\Sigma}_{\mathbf{x}} \mathbf{d} = \begin{pmatrix} A_D d_1 + B_D d_2 \\ B_D d_1 + C_D d_2 \\ A_H d_3 + B_H d_4 \\ B_H d_3 + C_H d_4 \\ u_v^2 d_5 \\ u_t^2 d_6 \end{pmatrix}. \quad (\text{A.22})$$

Each element of $\mathbf{w}$ is the scalar product of the corresponding row of the input covariance matrix and the gradient. The first element is $A_D d_1 + B_D d_2$, because the remaining elements of that covariance-matrix row are zero. Premultiplication by $\mathbf{d}^T$ gives the output variance.

$$u_c(F)^2 = \mathbf{d}^T \mathbf{w}, \quad (\text{A.23})$$

$$u_c(F)^2 = A_D d_1^2 + 2B_D d_1 d_2 + C_D d_2^2 + A_H d_3^2 + 2B_H d_3 d_4 + C_H d_4^2 + u_v^2 d_5^2 + u_t^2 d_6^2.$$

Symmetry of the population covariance matrices gives two equal covariance terms for each population. Their sums are $2B_D d_1 d_2$ and $2B_H d_3 d_4$ for the diseased and nondiseased populations, respectively. Cross-terms between independent sources are zero.

$$\begin{aligned} V_D &= A_D d_1^2 + 2B_D d_1 d_2 + C_D d_2^2, \\ V_H &= A_H d_3^2 + 2B_H d_3 d_4 + C_H d_4^2, \\ V_v &= u_v^2 d_5^2 = d_5^2 \frac{v(1-v)}{n_D + n_H}, \\ V_t &= u_t^2 d_6^2 = d_6^2 [b_0^2 + (b_1 t)^2], \\ u_c(F)^2 &= V_D + V_H + V_v + V_t. \end{aligned} \quad (\text{A.24})$$

Equation (A.24) decomposes the output variance into source-specific contributions and retains the covariance between the estimated mean and standard deviation within each population. Sampling variance includes the population-parameter contributions and, when included, the prevalence contribution. Measurement variance is the contribution propagated through the common threshold. The combined standard uncertainty is the nonnegative square root of their sum. The threshold expression in equation (A.24) uses the root-sum-of-squares model.

Within-population covariance terms may be positive or negative and are retained in the corresponding variance contributions. Each valid population covariance matrix yields a nonnegative propagated variance, although a covariance term may reduce that variance. Omitting a negative covariance term changes the assumed dependence structure.

For a generic measure F , combined, total sampling, and measurement standard uncertainties are denoted $u_c(F)$, $u_s(F)$, and $u_m(F)$, respectively; the parameter and prevalence sampling components are $u_{\text{par}}(F)$ and $u_{\text{prev}}(F)$. Figure legends use this notation, with the measure argument implicit where omitted.

The combined variance is the sum of the independent source-specific variance contributions, with all within-population covariances retained. Parameter sampling uncertainty comprises the diseased and nondiseased population contributions; total sampling uncertainty also includes the prevalence contribution.

$$u_s(F)^2 = u_{\text{par}}(F)^2 + u_{\text{prev}}(F)^2, \quad u_c(F)^2 = u_s(F)^2 + u_m(F)^2$$

A.7. Joint uncertainty propagation for all sixteen measures

$$\begin{aligned} V_a &= A_D a_m^2 + 2B_D a_m a_s + C_D a_s^2 + f_D(t)^2 u_t^2, \\ V_b &= A_H b_m^2 + 2B_H b_m b_s + C_H b_s^2 + f_H(t)^2 u_t^2, \\ C_{ab} &= -f_D(t) f_H(t) u_t^2. \end{aligned} \quad (\text{A.25})$$

The first three terms of V_a give the contribution from the estimated diseased population mean and standard deviation to the variance of Se; the final term gives the threshold contribution. V_b contains the corresponding nondiseased population and threshold contributions to the variance of Sp. Under the stated independence assumptions, C_{ab} arises solely from the common threshold and is nonpositive because the two threshold derivatives have opposite signs.

$$\begin{aligned} \Sigma_z &= \mathbf{K} \Sigma_x \mathbf{K}^T, \\ \Sigma_z &= \begin{pmatrix} V_a & C_{ab} & 0 \\ C_{ab} & V_b & 0 \\ 0 & 0 & u_v^2 \end{pmatrix}. \end{aligned} \quad (\text{A.26})$$

The intermediate covariance matrix has dimensions 3×3 . Covariances involving prevalence are zero under its assumed independence from the fitted population summaries and threshold. Other sampling designs may require additional covariance terms.

$$\begin{aligned} \Sigma_m &= \mathbf{M} \Sigma_x \mathbf{M}^T, \\ \Sigma_m &= \mathbf{G} \Sigma_z \mathbf{G}^T. \end{aligned} \quad (\text{A.27})$$

The two expressions yield the same 16×16 output covariance matrix because $\mathbf{M} = \mathbf{G} \mathbf{K}$. They represent equivalent formulations of uncertainty propagation. The factorization distinguishes the 16×3 Jacobian with respect to intermediate inputs from the 16×6 Jacobian with respect to primitive inputs.

$$\begin{aligned} \text{Cov}(F_i, F_j) &= G_{ia} V_a G_{ja} + G_{ia} C_{ab} G_{jb} + G_{ia} 0 G_{jv}, \\ &\quad + G_{ib} C_{ab} G_{ja} + G_{ib} V_b G_{jb} + G_{ib} 0 G_{jv}, \\ &\quad + G_{iv} 0 G_{ja} + G_{iv} 0 G_{jb} + G_{iv} u_v^2 G_{jv}, \\ \text{Cov}(F_i, F_j) &= G_{ia} V_a G_{ja} + G_{ia} C_{ab} G_{jb} + G_{ib} C_{ab} G_{ja} + G_{ib} V_b G_{jb} + G_{iv} u_v^2 G_{jv}. \end{aligned} \quad (\text{A.28})$$

The expanded product contains nine terms, four of which vanish under the assumed independence of prevalence. Off-diagonal elements are covariances between measures, and diagonal elements are their variances.

$$u_c(F_i)^2 = G_{ia}^2 V_a + 2 G_{ia} G_{ib} C_{ab} + G_{ib}^2 V_b + G_{iv}^2 u_v^2. \quad (\text{A.29})$$

The factor of two in equation (A.29) arises from the two equal covariance terms for Se and Sp. When C_{ab} is negative, their combined contribution decreases the propagated variance when the corresponding partial derivatives have the same sign and increases it when they have opposite signs. Omitting this contribution would represent threshold variation in Se and Sp as independent and would contradict the stated measurement model.

$$\begin{aligned} V_{t_i} &= u_t^2 (-G_{ia} f_D(t) + G_{ib} f_H(t))^2, \\ V_{t_i} &= u_t^2 [G_{ia}^2 f_D(t)^2 - 2 G_{ia} G_{ib} f_D(t) f_H(t) + G_{ib}^2 f_H(t)^2]. \end{aligned} \quad (\text{A.30})$$

Expansion of the squared derivative with respect to the common threshold recovers the covariance contribution for Se and Sp already included in the propagated variance.

A.7.1. Source-specific expansion in primitive coordinates

$$\Sigma_m = \mathbf{M}_D \Sigma_D \mathbf{M}_D^T + \mathbf{M}_H \Sigma_H \mathbf{M}_H^T + \mathbf{j}_v u_v^2 \mathbf{j}_v^T + \mathbf{j}_t u_t^2 \mathbf{j}_t^T. \quad (\text{A.31})$$

The diseased and nondiseased Jacobian submatrices comprise the first two and the next two columns of $\mathbf{M}$, respectively. The prevalence and threshold vectors $\mathbf{j}_v$ and $\mathbf{j}_t$ are its fifth and sixth columns. Each source-specific covariance contribution has dimensions 16×16 .

$$\begin{aligned}
\Sigma_{m_{ij}} = & A_D M_{i1} M_{j1} + B_D (M_{i1} M_{j2} + M_{i2} M_{j1}) + C_D M_{i2} M_{j2}, \\
& + A_H M_{i3} M_{j3} + B_H (M_{i3} M_{j4} + M_{i4} M_{j3}) + C_H M_{i4} M_{j4}, \\
& + u_v^2 M_{i5} M_{j5} + u_t^2 M_{i6} M_{j6}.
\end{aligned} \tag{A.32}$$

The expanded expression retains both within-population covariance terms. Equal output indices recover the scalar quadratic form, including the factors of two.

The diagonal elements of the output covariance matrix are the variances of the sixteen measures; the off-diagonal elements are their pairwise covariances. The scalar and joint formulations yield identical variances. Their agreement provides a check of implementation consistency but does not assess the fit of the population model to the observations.

A.8. Conversion of standard uncertainty to an approximate expanded-uncertainty interval

Standard uncertainty quantifies the scale of uncertainty, whereas expanded uncertainty defines an interval half-width. The coverage factor depends on the specified nominal coverage level and the approximate effective degrees of freedom. Each population's variance contribution retains the covariance between its estimated mean and standard deviation in the calculation of effective degrees of freedom.

The four nonnegative source-specific variance contributions are denoted V_D , V_H , V_v , and V_t , with the covariance between each population's estimated mean and standard deviation retained. Effective degrees of freedom are approximated by [20, 21]:

$$\begin{aligned}
V_D &= \nabla g_D^T \Sigma_D \nabla g_D, \\
V_H &= \nabla g_H^T \Sigma_H \nabla g_H, \\
V_v &= g_v^2 u_v^2, \\
V_t &= g_t^2 u_t^2. \\
\nu_{\text{eff}} &= \frac{(V_D + V_H + V_v + V_t)^2}{\frac{V_D^2}{\nu_D} + \frac{V_H^2}{\nu_H} + \frac{V_v^2}{\nu_v} + \frac{V_t^2}{\nu_t}}.
\end{aligned}$$

The sampling degrees of freedom are $\nu_D = n_D - 1$, $\nu_H = n_H - 1$, and $\nu_v = n_D + n_H - 1$. Assigning $\nu_t = \infty$ treats the threshold standard uncertainty as known without appreciable uncertainty. This assumption does not follow solely from classification as a Type B evaluation. For a nominal coverage level of $1 - \alpha$:

$$k = t_{\nu_{\text{eff}}, 1-\alpha/2}, \quad U = k u_c(F)$$

Here k is the $1 - \alpha/2$ quantile of Student's t distribution with ν_{eff} degrees of freedom, and U is the expanded uncertainty. Infinite threshold degrees of freedom remove the corresponding term from the denominator of the effective-degrees-of-freedom expression; the threshold contribution remains in the combined variance.

The two-sided expanded-uncertainty interval is:

$$I_{1-\alpha}(F) = [F - U, F + U]$$

At specified thresholds, these pointwise expanded-uncertainty intervals are based on a first-order Wald approximation. At optimized thresholds, measurement uncertainty is propagated nonlinearly by Gauss-Hermite quadrature, conditional on the estimated optimal threshold. The notebook permits restriction of interval endpoints to each measure's admissible range and retains the unrestricted intervals for verification. Restriction to the admissible range does not ensure nominal coverage or remove the limitations of the approximation. Standard uncertainties are unaffected by this restriction.

A.9. Evaluation of uncertainty at an estimated optimal threshold

In *DiagAccU.nb*, Wolfram NMaximize maximizes the selected measure's point estimate subject to $t > m_H$, without a prescribed upper threshold bound. The fitted population distributions remain fixed. At the

estimated optimal threshold, first-order sampling variance is combined with measurement variance obtained by nine-point Gauss–Hermite quadrature under a normal model for the common threshold.

The resulting interval quantifies uncertainty in the diagnostic accuracy measure conditional on the estimated optimal threshold subject to $t > m_H$. It excludes uncertainty associated with threshold estimation and selection.

At a smooth interior maximum, the first derivative with respect to the threshold is zero. First-order propagation may therefore yield a zero contribution from measurement uncertainty, although variation about the optimal threshold changes the measure. Nonlinear propagation accounts for these changes through weighted evaluations of the complete measure at quadrature nodes.

Under the normal threshold model, the random threshold is represented by the estimated optimal threshold plus its standard uncertainty multiplied by a standard normal variable. Nine-point Gauss–Hermite quadrature approximates expectations by nine weighted function evaluations. For nodes and weights defined with respect to the weight $\exp(-x^2)$, the transformed threshold nodes and normalized weights are:

$$t_i = t^* + \sqrt{2} u_t(t^*) x_i, \quad p_i = w_i / \sqrt{\pi}, \quad i = 1, \dots, 9.$$

The complete diagnostic accuracy measure is evaluated at each quadrature node, with the same threshold applied to both population distributions and the fitted population summaries and prevalence held fixed. The threshold standard uncertainty is evaluated at the estimated optimal threshold and held fixed throughout the integration. The weighted variance of the nine evaluations gives the measurement variance contribution. Its sum with the first-order sampling variance gives the combined variance, whose nonnegative square root is the combined standard uncertainty.

The estimated optimal threshold remains the center of the conditional analysis. The weighted mean used to calculate the measurement variance may differ from the measure evaluated at that threshold. Population distributions are not re-estimated, and threshold selection is not repeated. Assessment of threshold-selection uncertainty requires repetition of these estimation and selection steps, for example in an appropriately specified bootstrap. For positive-support families, threshold evaluations follow the support conventions of the fitted distributions.

A.10. Implementation and reproducibility

A single `AroundReplace` operation propagates the final scalar measure from a `VectorAround` object for the four correlated population summaries, an optional `Around` input for prevalence, and one `Around` input for the shared threshold [25, 26]. `DiagnosticAccuracyAllComponents`, `DiagnosticAccuracyAllConfidenceIntervals`, and `DiagnosticAccuracyJointResult` provide reproducible component, interval, and covariance records, respectively.

DiagAccU.nb is the computational source. *DiagAccUCalculations.nb* presents the symbolic calculations, and the *Interface Documentation* describes software controls and options. Sections A.4–A.7 integrate the expanded covariance products and all 48 intermediate [43–45].

Implementation checks comprise agreement between each scalar variance and the corresponding diagonal element of the joint output covariance matrix, agreement between the sum of the four source-specific covariance contributions and the total, symmetry, and nonnegative source-specific variances. These algebraic checks assess internal consistency. Interval coverage, model fit, transportability, and uncertainty associated with threshold selection require separate assessment.

Appendix B. Abbreviations, notation, and software

B.1. List of abbreviations

DAM: diagnostic accuracy measure

CI: confidence interval; the software label for an approximate pointwise expanded-uncertainty interval

GUM: Guide to the Expression of Uncertainty in Measurement

OGTT: oral glucose tolerance test

FPG: fasting plasma glucose

ROC: receiver operating characteristic

B.2. Notation

B.2.1. Populations

H: nondiseased population

D: diseased population

B.2.2. Test outcomes

TN: true negative

TP: true positive

FN: false negative

FP: false positive

B.2.3. DAMs

Se: sensitivity

Sp: specificity

PPV: positive predictive value

NPV: negative predictive value

ACC: overall diagnostic accuracy

DOR: diagnostic odds ratio

LR^+ : positive likelihood ratio

LR^- : negative likelihood ratio

J : Youden's index

BA: balanced accuracy

CZ: concordance probability

FMI: Fowlkes–Mallows index

κ : Cohen's kappa

PABAK: prevalence- and bias-adjusted kappa

F_1 : F_1 score

MCC: Matthews correlation coefficient

B.2.4. Parameters

μ_P, m_D, m_H : population arithmetic mean and supplied sample means, respectively; P denotes D or H

σ_P, s_D, s_H : population standard deviation and supplied Bessel-corrected sample standard deviations, respectively

n_D, n_H : sample sizes of the diseased and nondiseased populations, respectively

v : prevalence

t : diagnostic threshold

t^* : optimal threshold subject to $t > m_H$

$1 - \alpha$: nominal confidence level

B.2.5. Functions and relations

$u_s(F)$: total sampling standard uncertainty of measure F

$u_m(F)$: measurement standard uncertainty of F

$u_c(F)$: combined standard uncertainty of F

v_D, v_H, v_v, v_t : independent source-block variance contributions

$f_D(t), f_H(t)$: population-specific probability densities evaluated at the diagnostic threshold

$F_D(t), F_H(t)$: population-specific cumulative distribution functions evaluated at the diagnostic threshold

q : predicted-positive probability; p_o : observed agreement; p_e : chance agreement

Σ_x : covariance matrix of the six primitive inputs; J : output Jacobian

$I_{1-\alpha}(F)$: approximate expanded-uncertainty interval for F

$\text{Var}(F)$: variance of F ; $\text{Cov}(A, B)$: covariance of A and B

ν_{eff} : source-block effective degrees of freedom; k : coverage factor; $U = k u_c$: expanded uncertainty

B.3. Input

B.3.1. Range of input parameters

Most threshold controls use $t_{\min} = \max(0.01, \min(m_H - 6s_H, m_D - 6s_D))$ and $t_{\max} = \max(m_H + 6s_H, m_D + 6s_D)$. The selected-measure standard-uncertainty table across all distribution pairs (tdpu) instead uses $\max(m_H - 5s_H, 0) \leq t \leq m_D + 5s_H$.

n_D : 2 – 10,000

m_D : 0.1 – 10,000

s_D : 0.01 – 1,000

$n_{\bar{H}}$: 2 – 10,000

$m_{\bar{H}}$: 0.1 – 10,000

$s_{\bar{H}}$: 0.01 – 1,000

v : 0.001 – 0.999

n_U : 20–10,000. Interval views use this control to assign $\nu_t = n_U - 1$ to the threshold source. This control changes the threshold-source degrees of freedom and coverage factor; it does not change the threshold standard uncertainty. The illustrative calculations in Table 4 use $\nu_t = \infty$.

b_0 : $0 \leq b_0 \leq s_H$

b_1 : 0–0.1

$1 - \alpha$: 0.900–0.999

The threshold t , both population means m_D and m_H , both standard deviations s_D and s_H , and b_0 must use the same measurement unit; b_1 is dimensionless.

n : 20–20,000 in views with a total-sample-size control. In the sample-size profile, the total sample size is varied automatically while prevalence is held fixed at the selected value.

B.3.2. Additional input options

B.3.2.1. Plots

Extended and limited settings change the profile domain. Their limits depend on the selected view; they do not define the optimization domain or establish interval coverage.

B.3.2.2. Tables

Tables provide a decimal-display control from 1 to 10, with a default of 3. This changes displayed precision, not the precision of the underlying calculation.

B.3.3. Program controls

The program uses a tabbed interface for entering inputs and viewing estimates, uncertainty components, and plots.

Users may define the numerical settings with menus or sliders. Sliders are finely manipulated by pressing the *alt* or *opt* key while dragging the mouse. Pressing the *shift* or *ctrl* keys can even more finely manipulate them.

Plot interaction depends on the Wolfram Front End and the selected view. The interface documentation describes the available controls and profile domains.

B.4. Software availability and requirements

- a) Program name: *DiagAccU*
- b) Version: 2.0.0
- c) DOI: [10.5281/zenodo.22817814](https://doi.org/10.5281/zenodo.22817814)
- d) Project home page: <https://www.hcsl.com/Tools/DiagnosticAccuracy/>
- e) Computational source notebook: *DiagAccU.nb*. Available at:
- f) <https://www.hcsl.com/Tools/DiagnosticAccuracy/DiagAccU.nb>
- g) Programming language: Wolfram Language
- h) Operating systems: Microsoft Windows 10+, Linux 4.18+, Apple macOS 13+
- i) Hardware requirements: x86-64-v2 and higher or Apple Silicon microarchitecture CPU, and 16 GB+ RAM, recommended.
- j) Software citation: Chatzimichail RA, Chatzimichail T, Hatjimihail AT. *DiagAccU: a unified framework for uncertainty estimation of diagnostic accuracy measures*. Version 2.0.0. Hellenic Complex Systems Laboratory; 2026. doi: <https://doi.org/10.5281/zenodo.22817814>.
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